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Previous Release Version	5.0.0
AUTOSAR Specification	AUTOSAR Specification Release 4.0 Rev 3, Release 3.2 Rev1
Processor Platform	TC23x AB (with Sub-Derivatives)**

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In Chapter 1, all Released items are listed. The additional files listed in chapter 2 are demo software and for evaluation only.

Disclaimers:

- 1) The target of the delivery was ASIL B process compliance to ISO26262 for selected drivers***. However, Safety claim is not available for this release due to the following open FS topics:
 - a. Inconsistencies between the internal documents have been reported as part of the FS assessment. A detailed analysis with respect to Safety measures has been performed which indicates a low impact on the functionality, however requires a documentation change in the internal documents. The document update including traceability is ongoing.
 - b. Confirmation of test cases based on boundary and equivalence class is ongoing.
 - c. MCU driver also provides initialisation and interrupt handling services for the GTM peripheral. Peer reviews have been performed for the GTM driver artefacts but the FS assessment has indicated a need for the inspection type of reviews. An additional inspection review has been triggered and is ongoing.
 - d. The confirmation reviews for the changed FS artefacts is ongoing.

*** MCU, PORT, DIO, ADC, ICU, PWM, GPT, SPI, CRC and MCALLIB modules.

(**) Refer section 9 for supported Sub-Derivatives.

1 Released Items

Module Implementation Version placed in the title of table for each driver corresponds to the SW Major version, SW Minor Version and SW Patch version of the driver.

1.1 BASE Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_BASE_ReleasedItems.txt and MC-ISAR_AS3XX_AURIX_TC23X_BASE_ReleasedItems.txt

1.1.1 MCU Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V4.6.1)

File Name	AS403/ AS321	Description
Mcu.h	Both	Exports Driver functionality.
Mcu_Dbg.h	AS403	Exports Variables relevant for Debugging.
Mcu_Local.h	Both	Mcu functionality shared across MCU files
Mcu_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Mcu.c	Both	Implementation of Driver functionality.
Mcu_Crc.c	Both	Needed if CRC module is used for clock initialization. File can be excluded if CRC is not used
Mcu_Dma.c	Both	Needed if SPI or ADC with DMA or DMA complex module is used. File initializes the DMA clock
Mcu_Platform.c	Both	Implementation of clock functionality per platform derivate
Mcu_Ver.c	Both	Implementation of AUTOSAR specific functionality.
Aurix_MC-ISAR_UM_MCUDriver.pdf	Both	User Manual for MCU Configuration
Aurix_MC-ISAR_MCU_Clock_Calculator.xlsm	Both	Clock Calculation for MCU
Mcu.xdm	Both	Module parameter description (Tresos format)
McuPreConfiguration.xdm	Both	Module Pre Configuration (Tresos format)
Mcu.bmd	Both	Module parameter description
Mcu_Cfg.h	Both	Code Template
Mcu_PBCfg.c	Both	Code Template
Mcu_Bswmd.arxml	AS403	BSW Module description file
Mcu_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Mcu_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	User Manual Reference file
Gtm.h	Both	Exports GTM functionality
Gtm_Local.h	Both	GTM functionality shared across GTM files.
Gtm_Protect.h	Both	Included to define all first level indirection macros for

		Protected mode support (Supervisory access via OS calls).
Gtm.c	Both	Implementation of GTM functionality.
Gtm_Platform.c	Both	Implementation of GTM functionality for ADC connections and BRC configuration
Aurix_MC-ISAR_UM_GTMDriver.pdf	Both	User Manual for GTM driver
Gtm_Cfg.h	Both	Code Template
Gtm_PBCfg.c	Both	Code Template Post Build parameters
Gtm_LCcfg.c	Both	Code template link time parameters (Notifications)
Gtm.m	Both	Code Template
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.2 PORT Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V3.1.0)

File Name	AS403/ AS321	Description
Port.h	Both	Exports Driver functionality.
Port_Ver.h	Both	Exports Autosar specific Driver functionality.
Port_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Port.c	Both	Implementation of Driver functionality.
Aurix_MC-ISAR_UM_PORTDriver.pdf	Both	User Manual for PORT Configuration
Port.xdm	Both	Module parameter description (Tresos format)
Port_Bswmd.arxml	AS403	BSW Module description file
Port_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Port.bmd	Both	Module parameter description
Port_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	This file contains reference to User Manual
Port_Cfg.h	Both	Code Template
Port_PBCfg.c	Both	Code Template
Port.m	Both	Code Template
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.3 DIO Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V3.1.0)

File Name	AS403/ AS321	Description
Dio.h	Both	Exports Driver functionality.
Dio_Ver.h	Both	Exports Autosar specific Driver functionality.
Dio_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Dio.c	Both	Implementation of Driver functionality.
Dio_Ver.c	Both	Implementation of Autosar specific Driver functionality.
Aurix_MC-ISAR_UM_DioDriver.pdf	Both	User Manual for DIO Configuration
Dio.xdm	Both	Module parameter description (Tresos format)
Dio_Bswmd.arxml	AS403	BSW Module description file
Dio.bmd	Both	Module parameter description
Dio_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Dio_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	This file contains reference to User Manual
Dio_Cfg.h	Both	Code Template
Dio_PBCfg.c	AS403	Code Template
Dio_Lcfg.c	AS321	Code Template
Dio.m	Both	Code Template
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.4 ADC Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V3.2.1)

File Name	AS403/ AS321	Description
Adc.h	Both	Exports Driver functionality.
Adc_Dbg.h	AS403	Exports Variables relevant for Debugging
Adc_Utility.h	Both	ADC utility functions and Macros used in Adc.c and Adc_ConvHandle.c are exported.
Adc_ConvHandle.h	Both	Adc conversion handle sub functions are exported
Adc_HwHandle.h	Both	Adc hardware access sub functions are exported
Adc_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Adc.c	Both	Implementation of Driver functionality.
Adc_Ver.c	Both	Implementation of Autosar specific Driver functionality.
Adc_ConvHandle.c	Both	Implementation of ADC conversion handle sub functions

		of ADC driver.
Adc_HwHandle.c	Both	ADC hardware access sub functions of ADC driver.
Adc_Calibration.c	Both	ADC calibration related sub functions of ADC driver. Software calibration code available in the file is not productive.
Aurix_MC-ISAR_UM_ADCDriver.pdf	Both	User Manual for ADC
Adc.xdm	Both	Module parameter description (Tresos format)
Adc.bmd	Both	Module parameter description
Adc_Bswmd.arxml	AS403	BSW Module description file
Adc_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Adc_Cfg.h	Both	Code Template
Adc_PBCfg.c	Both	Code Template
Adc.m	Both	Code Template
Adc_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.5 GPT Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V3.3.0)

File Name	AS403/ AS321	Description
Gpt.h	Both	Exports Driver functionality
Gpt_Dbg.h	AS403	Exports Variables relevant for Debugging
Gpt_Local.h	Both	GPT functionality shared across GPT files
Gpt_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Gpt.c	Both	Implementation of Driver functionality
Gpt_Ver.c	Both	Implementation of Autosar specific Driver functionality
Aurix_MC-ISAR_UM_GPTDriver.pdf	Both	User Manual for GPT
Gpt.xdm	Both	Module parameter description (Tresos format)
Gpt.bmd	Both	Module parameter description
Gpt_Bswmd.arxml	AS403	BSW Module description file
Gpt_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Gpt_Cfg.h	Both	Code Template

Gpt_PBCfg.c	Both	Code Template
Gpt.m	Both	Code Template
Gpt_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.6 ICU Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V3.11.0)

File Name	AS403/ AS321	Description
Icu_17_GtmCcu6.h	Both	Exports Driver functionality
Icu_17_GtmCcu6_Dbg.h	AS403	Exports Variables relevant for Debugging
Icu_17_GtmCcu6_Local.h	Both	ICU functionality shared across ICU files
Icu_17_GtmCcu6_Safe.h	Both	Exports Driver functionality specific to safety
Icu_17_GtmCcu6_Ver.h	Both	Exports Driver functionality specific to Autosar
Icu_17_GtmCcu6_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Icu_17_GtmCcu6.c	Both	Implementation of Driver functionality specific to safety
Icu_17_GtmCcu6_Safe.c	Both	Implementation of Driver functionality specific to Autosar
Icu_17_GtmCcu6_Ver.c	Both	Implementation of Driver functionality
Aurix_MC-ISAR_UM_ICUDriver.pdf	Both	User Manual for ICU
Icu_17_GtmCcu6.xdm	Both	Module parameter description (Tresos format)
Icu_17_GtmCcu6.bmd	Both	Module parameter description
Icu_17_GtmCcu6_Bswmd.arxml	AS403	BSW Module description file
Icu_17_GtmCcu6_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Icu_17_GtmCcu6_Cfg.h	Both	Code Template
Icu_17_GtmCcu6_PBCfg.c	Both	Code Template
Icu_17_GtmCcu6.m	Both	Code Template
Icu_17_GtmCcu6_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.7 PWM Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V5.1.0)

File Name	AS403/ AS321	Description
Pwm_17_Gtm.h	Both	Exports Driver functionality.
Pwm_17_Gtm_Ver.h	Both	Exports Driver functionality specific to Autosar.
Pwm_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Pwm_17_Gtm.c	Both	Implementation of Driver functionality.
Pwm_17_Gtm_Ver.c	Both	Implementation of Autosar specific Driver functionality
Aurix_MC-ISAR_UM_PWMDriver.pdf	Both	User Manual for PWM
Pwm_17_Gtm.xdm	Both	Module parameter description (Tresos format)
Pwm_17_Gtm.bmd	Both	Module parameter description
Pwm_17_Gtm_Bswmd.arxml	AS403	BSW Module description file
Pwm_17_Gtm_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Pwm_17_Gtm_Cfg.h	Both	Code Template
Pwm_17_Gtm_PBCfg.c	Both	Code Template
Pwm_17_Gtm.m	Both	Code Template
Pwm_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.8 SPIHandler Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V4.8.0)

File Name	AS403/ AS321	Description
Spi.h	Both	Exports Driver functionality.
Spi_Dbg.h	AS403	Exports Variables relevant for Debugging
Spi_Local.h	Both	SPI functionality shared across SPI files.
SpiSlave.h	Both	SPI functionalities and types specific to Slave.
Spi_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Spi.c	Both	Implementation of Driver functionality with respect to Master.
SpiSlave.c	Both	Implementation of Driver functionality with respect to Slave.
Spi_Ver.c	Both	Implementation of Driver functionality specific to Autosar.

Aurix_MC-ISAR_UM_SPIHandlerDriver.pdf	Both	User Manual for SPI
Spi.xdm	Both	Module parameter description (Tresos format)
Spi.bmd	Both	Module parameter description
Spi_Bswmd.arxml	AS403	BSW Module description file
Spi_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Spi_Cfg.h	Both	Code Template
Spi_PBCfg.c	Both	Code Template
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.1.9 WDG Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V3.4.0)

File Name	AS403/ AS321	Description
Wdg_17_Scu.h	Both	Exports Driver functionality.
Wdg_17_Scu_Dbg.h	AS403	Exports Variables relevant for Debugging
Wdg_17_Scu_Cbk.h	AS403	Exports call back functionality of WDG
Wdg_17_Scu_Safe.h	Both	Exports Driver functionality specific to safety
Wdg_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Wdg_17_Scu.c	Both	Implementation of Driver functionality
Wdg_17_Scu_Safe.c	Both	Implementation of Driver functionality specific to safety
Aurix_MC-ISAR_UM_WdgDriver.pdf	Both	User Manual for WDG
Wdg_17_Scu.xdm	Both	Module parameter description (Tresos format)
Wdg_17_Scu.bmd	Both	Module parameter description
Wdg_17_Scu_Bswmd.arxml	AS403	BSW Module description file
Wdg_17_Scu_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
Wdg_17_Scu_Cfg.h	Both	Code Template
Wdg_17_Scu_PBCfg.c	Both	Code Template
Wdg_17_Scu.m	Both	Code Template
Wdg_ConfigDoc.html	Both	Reference file to know SFR mapping per API and MPU settings for the driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file

MANIFEST.MF	Both	Tresos manifest file
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1.1.10 MCAL Library

File Name	AS403/ AS321	Description
Mcal_TcLib.c	Both	Implementation of - Global Functions: Tricore library functions.
Mcal_TcLib.h	Both	Export global functions used by all MCAL modules.
Mcal_WdgLib.c	Both	Implementation of - Global Functions: SCU library functions.
Mcal_WdgLib.h	Both	Export of SCU library functions.
Mcal_DmaLib.c	Both	Implementation of - Global Functions: DMA library functions.
Mcal_DmaLib.h	Both	Export of DMA library functions
Mcal_Compiler.h	Both	MCAL specific compiler abstraction is place here.
MC-ISAR_AURIX_UM_MCALLib.pdf	Both	User Manual for MCALLib

1.1.11 Register Header Files

File Name	AS403/ AS321	Description
lfx<Mod>_regdef.h	Both	Microcontroller's special function register declaration is located here. Mcal code is compliant to this Sfr file and hence this needs to be mandatorily used. Location: \\Aurix_MC-ISAR\general\tricore\inc\TC23x\
lfx<Mod>_reg.h	Both	Microcontroller's SFR Addresses are mentioned here. Mcal code is compliant to this Sfr file and hence this needs to be mandatorily used. Location: \\Aurix_MC-ISAR\general\tricore\inc\TC23x\

Note:

1. <Mod > refers to Asclin, Bmu, Cbs, Ccu6, Cif, Cpu, Dma, Ebcu, Emem, Eray, Eth, Gpt12, Gtm, Hsct, Int, Iom, Lmu, Can, Flash, Port, Qspi, Scu, Sdma, Vadc and Xbar.

2. Qualified SFR files version:

TC23x-AB: TC23xADAS SFR C Reference Header Package [29 Jul 2016 | V1.1.R1]

1.1.12 Common Header Files

Common header files included by all modules are mentioned below.

File Name	AS403/ AS321	Description
Mcal.h	Both	Export global functions used by all MCAL modules. Location: \\Aurix_MC-ISAR\\tricore_general\\ssc\\inc
Mcal_Options.h	Both	Defines the macros with respect to integration of MCAL and SafeTlib to include required code only. Location: \\Aurix_MC-ISAR\\tricore_general\\ssc\\inc
Std_Types.h	Both	Data types definition for the controller. Location: \\Aurix_MC-ISAR\\general\\inc
Compiler.h	Both	Compiler header file. Location: Aurix_MC-ISAR\\general\\tricore\\compiler
Compiler_Cfg.h	Both	Compiler abstraction is located here. Location: Aurix_MC-ISAR\\general\\tricore\\compiler
Platform_Types.h	Both	Data types definition for the controller. \\Aurix_MC-ISAR\\general\\tricore\\inc
SafetyReport.h	Both	Contains the prototype of Safety Error report function Location: \\Aurix_MC-ISAR\\integration_general\\inc

Note: Qualified code inside the specified files shall not be changed / modified. However customer can add extra code based on the need of application.

1.1.13 Tresos signature/property files

Signature files for all the modules are mentioned below.

File Name	Description
CRYPTOMANIFEST.MF CRYPTOMANIFESTSIG.MF	Signature files for a module Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\ <Mod>_Aurix\META-INF\
tresos_Partner_Infineon.lic	License for tresos
Resource.xdm	Tresos format file for selecting the sub-derivative
plugin.xml	Tresos plugin file for selecting the sub-derivative
AURIX_TC233.properties	Parameter Description (Tresos format) for TC233
AURIX_TC234.properties	Parameter Description (Tresos format) for TC234
AURIX_TC237.properties	Parameter Description (Tresos format) for TC237
AURIX_TC233L_16F200F.properties	Parameter Description (Tresos format) for TC233L_16F200F
AURIX_TC233L_32F200F.properties	Parameter Description (Tresos format) for TC233L_32F200F
AURIX_TC233LC_24F133F.properties	Parameter Description (Tresos format) for TC233LC_24F133F
AURIX_TC233LP_16F200F.properties	Parameter Description (Tresos format) for TC233LP_16F200F
AURIX_TC233LP_32F200F.properties	Parameter Description (Tresos format) for TC233LP_32F200F
AURIX_TC234L_32F200F.properties	Parameter Description (Tresos format) for TC234L_32F200F
AURIX_TC234LA_32F200F.properties	Parameter Description (Tresos format) for TC234LA_32F200F
AURIX_TC234LP_16F200F.properties	Parameter Description (Tresos format) for TC234LP_16F200F
AURIX_TC234LP_32F200F.properties	Parameter Description (Tresos format) for TC234LP_32F200F
AURIX_TC234LX_32F200F.properties	Parameter Description (Tresos format) for TC234LX_32F200F
AURIX_TC237L_32F200S.properties	Parameter Description (Tresos format) for TC237L_32F200S
AURIX_TC237LP_32F200S.properties	Parameter Description (Tresos format) for TC237LP_32F200S
plugin.properties	Tresos property file
MANIFEST.MF	Tresos manifest file

<Mod> refers to Mcu, Port, Dio, Gtm, Adc, Gpt, Icu_17_GtmCcu6, Pwm_17_Gtm, Spi, Wdg_17_Scu, Can_17_MCanP, Lin_17_AscLin, Fr_17_Eray, Eth_17_EthMac, FlsLoader, Uart, Dma, StdLin, Sent, Stm, Smu, Iom, Dem, Irq, and EcuM

1.2 COM_BASIC Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_COM_BASIC_ReleasedItems.txt and MC-ISAR_AS3XX_AURIX_TC23X_COM_BASIC_ReleasedItems.txt

1.2.1 LIN Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V 2.3.1)

File Name	AS403/ AS321	Description
Lin_17_AscLin.h	Both	Exports Driver functionality.
Lin_17_AscLin.c	Both	Implementation of Driver functionality.
Lin_17_AscLin_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_LINDriver.pdf	Both	User Manual for LIN
Lin_17_AscLin.xdm	Both	Module parameter description (Tresos format).
Lin_17_AscLin_Cfg.h	Both	Code Template
Lin_17_AscLin_PBCfg.c	Both	Code Template
Lin_17_AscLin.bmd	Both	Module parameter description (AUTOSAR format).
Lin_17_AscLin_Bswmd.arxml	AS403	BSWMD file
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.2.2 CAN Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V 2.7.1)

File Name	AS403/ AS321	Description
Can_17_MCanP.h	Both	Exports Driver functionality.
Can_17_MCanP.c	Both	Implementation of Driver functionality.
Can_17_MCanP_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_CANDriver.pdf	Both	User Manual for CAN
Can_17_MCanP.xdm	Both	Module parameter description (Tresos format).
Can_17_MCanP_Cfg.h	Both	Code Template
Can_17_MCanP_PBCfg.c	Both	Code Template
Can_17_MCanP.m	Both	Code Template
Can_17_MCanP.bmd	Both	Module parameter description (AUTOSAR format).
Can_17_MCanP_Bswmd.arxml	AS403	BSWMD file

anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.2.3 CanTrcv6251G Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Pre Compile Variant) (Module Implementation Version V 2.2.1)

File Name	AS403/ AS321	Description
CanTrcv_17_6251G.h	Both	Exports Driver functionality.
CanTrcv_17_6251G.c	Both	Implementation of Driver functionality.
Aurix_MC-ISAR_UM_ CanTrcv_17_6251GDriver.pdf	Both	User Manual for CAN Transceiver
CanTrcv_17_6251G.xdm	Both	Module parameter description (Tresos format).
CanTrcv_17_6251G_Cfg.h	Both	Code Template
CanTrcv_17_6251G_Cfg.c	Both	Code Template
CanTrcv_17_6251G.bmd	Both	Module parameter description (AUTOSAR format).
CanTrcv_17_6251G_Bswmd.arxml	AS403	BSWMD file
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.2.4 CanTrcv6250GV33 Driver (AUTOSAR Release 4.0 Rev3, Pre Compile Variant) (Module Implementation Version V 2.1.8 for AS403, V 2.1.7 for AS321)

File Name	AS403/ AS321	Description
CanTrcv_17_6250GV33.h	Both	Exports Driver functionality
CanTrcv_17_6250GV33.c	Both	Implementation of Driver functionality.
Aurix_MC-ISAR_UM_ CanTrcv_17_6250GV33Driver.pdf	Both	User Manual for CAN
CanTrcv_17_6250GV33.xdm	Both	Module parameter description (Tresos format).
CanTrcv_17_6250GV33_Cfg.h	Both	Code Template
CanTrcv_17_6250GV33_Cfg.c	Both	Code Template
CanTrcv_17_6250GV33.bmd	Both	Module parameter description (AUTOSAR format).
anchors.xml	Both	User Manual Reference file
CanTrcv_17_6250GV33_Bswmd. arxml	AS403	BSWMD file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file

MANIFEST.MF	Both	Tresos manifest file
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1.3 COM_ENHANCED Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_COM_ENHANCED_ReleasedItems.txt and MC-ISAR_AS3XX_AURIX_TC23X_COM_ENHANCED_ReleasedItems.txt

1.3.1 FlexRay Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V 2.4.1)

File Name	AS403/ AS321	Description
Fr_17_Eray.h	Both	Exports Driver functionality.
Fr_17_Eray_Dbg.h	AS403	Exports Variables relevant for Debugging.
Fr_17_Eray_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Fr_17_Eray.c	Both	Implementation of Driver functionality
Aurix_MC-ISAR_UM_FR_17_ERAYDriver.pdf	Both	User Manual for Flex ray
Fr_17_Eray.xdm	Both	Module parameter description (Tresos format).
Fr_17_Eray.bmd	Both	Module parameter description (AUTOSAR format).
Fr_17_Eray_Cfg.h	Both	Code Template
Fr_17_Eray_PBCfg.c	Both	Code Template
Fr_17_Eray.m	Both	Code Template
FlexTrain.exe	Both	Tool to generate LPDU index to message buffer index mapping. This tool is automatically executed through the batch file to update the configuration file with LPDU configuration data. Tool is used only if message buffer reconfiguration is required. For usage instructions, refer Readme_FlexTrain.txt file available in the same folder
Fr_17_Eray_Bswmd.arxml	AS403	BSWMD file
Fr_17_Eray_Catalog.xml	AS403	XML Data Model for FR driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos property file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.3.2 Ethernet Driver (AUTOSAR Release 4.0 Rev3, Post Build Variant) (Module Implementation Version V 4.3.2)

File Name	AS403/ AS321	Description
Eth_17_EthMac.h	AS403	Exports Driver functionality.
Eth_17_EthMac_Types.h	AS403	Exports Driver functionality.

Eth_17_EthMac_Dbg.h	AS403	Exports Driver functionality.
Eth_17_EthMac_Protect.h	AS403	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Eth_17_EthMac.c	AS403	Implementation of Driver functionality
Aurix_MC-ISAR_UM_ETHDriver.pdf	AS403	User Manual for Ethernet Driver
Eth_17_EthMac.xdm	AS403	Module parameter description (Tresos format).
Eth_17_EthMac.bmd	AS403	Module parameter description (AUTOSAR format).
Eth_17_EthMac_Cfg.h	AS403	Code Template
Eth_17_EthMac_PBCfg.c	AS403	Code Template
Eth_17_EthMac_Bswmd.arxml	AS403	BSWMD file
Eth_17_EthMac_Catalog.xml	AS403	XML Data Model for ETH driver
anchors.xml	AS403	User Manual Reference file
plugin.xml	AS403	Tresos property file
plugin.properties	AS403	Tresos property file
MANIFEST.MF	AS403	Tresos manifest file

1.4 MEM Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_MEM_ReleasedItems.txt and MC-ISAR_AS4XX_AURIX_TC23X_MEM_ReleasedItems.txt

1.4.1 FLS Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Post Build Variant) (Module Implementation Version V2.4.0)

File Name	AS403/ AS321	Description
Fls_17_Pmu.h	Both	Exports Driver functionality.
Fls_17_Pmu.c	Both	Implementation of Driver functionality.
Fls_17_Pmu_Ver.c	Both	Implementation of AUTOSAR version specific functionality of FLS driver.
Fls_17_Pmu_Local.h	Both	FLS_17_PMU driver internal header file
Fls_17_Pmu_ac.h	Both	This file contains command cycles of Flash
Fls_17_Pmu_Dbg.h	AS403	Exports Debug support functionality.
Fls_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Fls_17_Pmu_ac.c	Both	This file contains command cycles of Flash driver
Aurix_MC-ISAR_UM_FLS_17_PMUDriver.pdf	Both	User Manual for FLS
Fls_17_Pmu.xdm	Both	Module parameter description (Tresos format).
Fls_17_Pmu_Cfg.h	Both	Code Template

Fls_17_Pmu_PBCfg.c	Both	Code Template
Fls_17_Pmu.bmd	Both	Module parameter description (AUTOSAR format).
Fls_17_Pmu_Bswmd.arxml	AS403	BSW Module description file
Fls_17_Pmu_Catalog.xml	AS403	XML Data Model for FLS driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.4.2 FEE Driver (AUTOSAR Release 4.0 Rev3 and 3.2 Rev1, Pre Compile Variant) (Module Implementation Version V2.4.0)

File Name	AS403/ AS321	Description
Fee.h	Both	Exports Driver functionality.
Fee.c	Both	Implementation of Driver functionality.
Fee_Ver.c	Both	Implementation of AUTOSAR version specific functionality of FEE driver.
Fee_Cbk.h	Both	Exports Notification functionality.
Fee_Dbg.h	AS403	Exports Debug functionality.
Aurix_MC-ISAR_UM_FEEDriver.pdf	Both	User Manual for FEE.
Fee.xdm	Both	Module parameter description (Tresos format).
Fee_Cfg.h	Both	Code Template
Fee_PBCfg.c	Both	Code Template
Fee.bmd	Both	Module parameter description (AUTOSAR format).
Fee_Bswmd.arxml	AS403	BSWMD file
Fee_Catalog.xml	AS403	XML Data Model for FEE driver
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
Aurix_MC-ISAR_FEE_Generate_StatePage.exe	Both	Initial State Page generation tool (demo version)
Aurix_MC-ISAR_FEE_Generate_StatePage_ReadMe.txt	Both	ReadMe file for Initial State Page generation tool
Aurix_MC-ISAR_FEE_RetentionCalculator.xlsx	Both	FEE Retention calculator (demo version)
Aurix_MC-ISAR_FEE_Generate_InitialData.pl	Both	Initial state page and data block hex generation tool
Aurix_MC-ISAR_FEE_Generate_InitialData_ReadMe.txt	Both	ReadMe file for initial state page and data block hex generation tool

plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.5 LIB Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_LIB_ReleasedItems.txt and MC-ISAR_AS3XX_AURIX_TC23X_LIB_ReleasedItems.txt

1.5.1 CRC Library (AUTOSAR Release 4.0 Rev 3 and 3.2 Rev1, Pre Compile Variant) (Module Implementation Version V3.2.0)

File Name	AS403/ AS321	Description
Crc.h	Both	Exports Library functionality.
Crc.c	Both	Implementation of Library functionality.
Crc_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_CRCLibrary.pdf	Both	User Manual for CRC
Crc.xdm	Both	Module parameter description (AUTOSAR format).
Crc_Cfg.h	Both	Code Template
Crc.bmd	Both	Module parameter description (AUTOSAR format).
Crc_Bswmd.arxml	AS403	BSWMD File
Crc_Catalog.xml	AS403	Catalog File
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.5.2 BFX Library (AUTOSAR Release 4.0 Rev 3, Pre Compile Variant) (Module Implementation Version V3.0.0)

File Name	AS403/ AS321	Description
Bfx.h	AS403	Exports Library functionality.
Bfx.c	AS403	Implementation of Library functionality.
Aurix_MC-ISAR_UM_BFXLibrary.pdf	AS403	User Manual for BFX
Bfx_Bswmd.arxml	AS403	BSWMD File
Bfx_Catalog.xml	AS403	Catalog File
anchors.xml	AS403	User Manual Reference file
plugin.properties	AS403	Tresos property file

1.6 COMPLEX Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_COMPLEX_ReleasedItems.txt and MC-ISAR_AS4XX_AURIX_TC23X_COMPLEX_ReleasedItems.txt

1.6.1 FlsLoader Driver (Complex Driver, Pre compile Variant) (Module Implementation Version V2.2.9)

File Name	AS403/ AS321	Description
FlsLoader.h	Both	Exports Driver functionality.
FlsLoader.c	Both	Implementation of Driver functionality.
FlsLoader_Platform.c	Both	Implementation of Flsloader functionality for Platform derivative
FlsLoader_Local.h	Both	Flsloader functionality shared across Flsloader files
Aurix_MC-ISAR_UM_FlsLoaderDriver.pdf	Both	User Manual for Flsloader
FlsLoader.xdm	Both	Module parameter description (AUTOSAR format).
FlsLoader.bmd	Both	Module parameter description (AUTOSAR format).
FlsLoader_Cfg.h	Both	Code Template
anchors.xml	Both	User manual path file
FlsLoader_Bswmd.arxml	AS403	BSW Module description file
FlsLoader_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.6.2 UART Driver (Complex Driver, Post Build Variant) (Module Implementation Version V 1.3.0)

File Name	AS403/ AS321	Description
Uart.h	Both	Exports Driver functionality.
Uart.c	Both	Implementation of Driver functionality.
Uart_Dbg.h	AS403	Exports Debug variables.
Uart_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_UARTDriver.pdf	Both	User Manual for UART driver
Uart.xdm	Both	Module parameter description (AUTOSAR format).
Uart.bmd	Both	Module parameter description (AUTOSAR format).
Uart_PBCfg.c	Both	Code Template
Uart_Cfg.h	Both	Code Template
anchors.xml	Both	User manual path file
Uart_Bswmd.arxml	AS403	BSW Module description file

Uart_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.6.3 DMA Driver (Complex Driver, Post Build Variant) (Module Implementation Version V 1.2.0)

File Name	AS403/ AS321	Description
Dma.h	Both	Exports Driver functionality.
Dma.c	Both	Implementation of Driver functionality.
Dma_Dbg.h	AS403	Exports Debug variables.
Dma_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_DmaDriver.pdf	Both	User Manual for DMA driver
Dma.xdm	Both	Module parameter description (AUTOSAR format).
Dma.bmd	Both	Module parameter description (AUTOSAR format).
Dma_PBCfg.c	Both	Code Template
Dma_Cfg.h	Both	Code Template
anchors.xml	Both	User manual path file
Dma_Bswmd.arxml	AS403	BSW Module description file
Dma_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.7 STDLIN_COMPLEX Package:

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_STDLIN_COMPLEX_ReleasedItems.txt and MC-ISAR_AS4XX_AURIX_TC23X_STDLIN_COMPLEX_ReleasedItems.txt

1.7.1 StdLin Driver (Complex Driver, Pre Compile Variant) (Module Implementation Version V 1.1.0)

File Name	AS403/ AS321	Description
StdLin.h	Both	Exports Driver functionality.
StdLin.c	Both	Implementation of Driver functionality.
StdLin_Hal.c	Both	Implementation of Hw dependent functions
StdLin_Hal.h	Both	Header file for Hw dependent Layer
StdLin_Protocol.c	Both	Implementation of protocol and scheduler functionality

StdLin_Protocol.h	Both	Header file for protocol layer
StdLin_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
StdLin_TpDiag.c	Both	Implementation of Transport protocol and diagnostic functionality
StdLin_TpDiag.h	Both	Header file for TPDia layer
StdLin_Types.h	Both	Header file for common type definitions
Aurix_MC-ISAR_UM_StdLinDriver.doc	Both	User Manual for StdLin
StdLin.xdm	Both	Module parameter description (AUTOSAR format).
StdLin.bmd	Both	Module parameter description (AUTOSAR format).
StdLin_Cfg.h	Both	Code Template
StdLin_Cfg.c	Both	Code Template
StdLin_Bswmd.arxml	AS403	BSW Module description file
StdLin_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file.
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.8 DEMOCDD Package (This package is Alpha quality):

All the released files location and file version number of TC23xAB device is available in MC-ISAR_AS4XX_AURIX_TC23X_DEMOCDD_ReleasedItems.txt and MC-ISAR_AS4XX_AURIX_TC23X_DEMOCDD_ReleasedItems.txt

1.8.1 SENT Driver (Complex Driver, Post Build Variant) (Module Implementation Version V 0.2.0)

File Name	AS403/ AS321	Description
Sent.c	Both	Implementation of Driver functionality.
Sent.h	Both	Exports Driver functionality.
Sent_Types.h	Both	Contains Typedefs required for SENT module
Sent_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_SENTDriver.doc	Both	User Manual for Sent driver
Sent.bmd	Both	Module parameter description (AUTOSAR format).
Sent_Cfg.h	Both	Code Template
Sent_LCf.c	Both	Code Template
Sent_PBf.c	Both	Code Template

anchors.xml	Both	User manual path file
Sent_Bswmd.arxml	AS403	BSW Module description file
Sent_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.8.2 STM Driver (Complex Driver, Post Build Variant) (Module Implementation Version V 0.0.1)

File Name	AS403/ AS321	Description
Stm.h	Both	Exports Driver functionality.
Stm.c	Both	Implementation of Driver functionality.
Stm_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_STMDriver.doc	Both	User Manual for STM driver
anchors.xml	Both	User manual path file
Stm_Bswmd.arxml	AS403	BSW Module description file
Stm_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
MANIFEST.MF	Both	Tresos manifest file

1.8.3 SMU Driver (Complex Driver, Post Build Variant) (Module Implementation Version V 0.0.1)

File Name	AS403/ AS321	Description
Smu.h	Both	Exports Driver functionality.
Smu.c	Both	Implementation of Driver functionality.
Aurix_MC-ISAR_UM_SmuDriver.doc	Both	User Manual for SMU driver
Smu.xdm	Both	Module parameter description (AUTOSAR format).
Smu.bmd	Both	Module parameter description (AUTOSAR format).
Smu_PBCfg.c	Both	Code Template
Smu_Cfg.h	Both	Code Template
Smu.m	Both	Code Template
Ifx_MemMap.h	Both	Mapping of Memory section required for SMU
Mcal_Lib.h	Both	Mcal library declarations required for SMU
anchors.xml	Both	User manual path file

plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

1.8.4 IOM Driver (Complex Driver, Post Build Variant) (Module Implementation Version V 2.1.0)

File Name	AS403/ AS321	Description
lom.h	Both	Exports Driver functionality.
lom.c	Both	Implementation of Driver functionality.
lom_Protect.h	Both	Included to define all first level indirection macros for Protected mode support (Supervisory access via OS calls).
Aurix_MC-ISAR_UM_lomDriver.doc	Both	User Manual for IOM driver
lom.xdm	Both	Module parameter description (AUTOSAR format).
lom.bmd	Both	Module parameter description (AUTOSAR format).
lom_PBCfg.c	Both	Code Template
lom_Cfg.h	Both	Code Template
anchors.xml	Both	User manual path file
lom_Bswmd.arxml	AS403	BSW Module description file
lom_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

2 MCAL Support Packages

The following artifacts are given for Evaluation Purpose only.

2.1 Demo Software

File Name	AS403/ AS321	Description
Mcal.c	Both	Implementation of - Sample Start up Code. Location: \\Aurix_MC-ISAR\tricare_general\ssc\src
\\Aurix_MC-ISAR\<mod>_infineon_tricare\demo\src\<mod>_Demo.c	Both	Demo files for the respective modules
\\Aurix_MC-ISAR\<mod>_infineon_tricare\demo\inc\<mod>_Demo.h	Both	Respective header files for demo.
\\Aurix_MC-ISAR\fr_17_era_y_infineon_tricare\tool\HeaderCRC.exe \\Aurix_MC-ISAR\fr_17_era_y_infineon_tricare\tool\HeaderCRC.c \\Aurix_MC-ISAR\fr_17_era_y_infineon_tricare\tool\readme.txt \\Aurix_MC-ISAR\fr_17_era_y_infineon_tricare\tool\Readme_FlexTrain.txt	Both	Support files to generate Header CRC
\\Aurix_MC-ISAR\<mod>_infineon_tricare\demo\src\StdLin_Utility.c	Both	utility file for stdlin demo
\\Aurix_MC-ISAR\<mod>_infineon_tricare\demo\inc\StdLin_Utility.h	Both	Utility header file for stdlin demo
\\Aurix_MC-ISAR\<mod>_infineon_tricare\demo\doc\DemoApp.ldf	Both	LDF file used for the stdlin demo
\\Aurix_MC-ISAR\demoapp\src\DemoApp.c	Both	DemoApp application file, that calls demo functions
\\Aurix_MC-ISAR\demoapp\src\DemoApp_FctCall.ct	Both	Function template used for Call by function functionality of DemoApp.
\\Aurix_MC-ISAR\demoapp\inc\DemoApp_FctCall.h	Both	DemoApp application file.
DemoApp	Both	Demo files for the respective modules for Selectable. Location: a) \\Aurix_MC-ISAR\demoapp\cfg\tresos_workspace\Selectable_Cfg b) \\Aurix_MC-ISAR\demoapp\cfg\tresos_workspace\Loadable_Cfg

DemoApp_Cfg.h, ModuleDelivery.bat	Both	Generated in EcuM for the DemoApp purpose.
CopyTresosPlugins.bat, DemoAppEpcCodeGen.bat, DemoAppNoCfgSelectable.bat	Both	Files to support Demo application Location: \\Aurix_MC-ISAR\demoapp\Selectable_bin
DemoAppClean.bat, DemoAppStart.bat, DemoAppWipe.bat, MCAL_OpFileUpdate.bat	Both	Files to support Demo application Location: a) \\Aurix_MC-ISAR\demoapp\Loadable_bin b) \\Aurix_MC-ISAR\demoapp\LoadableCfg_bin \\Aurix_MC-ISAR\demoapp\Selectable_bin
get_functions_dwarf.pl	Both	Perl script used for Call by Function functionality of DemoApp. Location: \\Aurix_MC-ISAR\demoapp\Selectable_bin_scripts \\Aurix_MC-ISAR\demoapp\Loadable_bin_scripts
GNU contents	Both	Files for executing GNU make.

<Mod> refers to Mcu, Port, Dio, Gtm, Adc, Gpt, Icu_17_GtmCcu6, Pwm_17_Gtm, Spi, Wdg_17_Scu, Can_17_MCanP, Lin_17_AscLin, Fr_17_Eray, Eth_17_EthMac, FlsLoader, Uart, Dma, StdLin, Sent, Stm, Smu and Iom.

2.2 Test Software Package

Package	AS403/ AS321	Description
Integration files for Demo Software	Both	Sample Demo Application Package, Refer: Demo_App.txt for details of the Application Location: \\Aurix_MC-ISAR\demoapp General Test Files, Test stubs for: DET Functions (Det.c, Det.h) Safety Error Callback functions (SafetyReport.c) DEM Functions (Dem.c, Dem.h) DEM Configuration (Dem.xdm, Dem_PBCfg.h) OS Functions (Os.c, Os.h) Fls callback API (Fls_17_Cbk.h) Interface files Location: a) \\Aurix_MC-ISAR\integration_general b) \\Aurix_MC-ISAR\tools\tresosECU\plugins\ Dem_Aurix\config c) \\Aurix_MC-ISAR\tools\tresosECU\plugins\ Dem_Aurix\generate\inc
MemMap.h	Both	Memory Sections are defined here. Location: Aurix_MC-ISAR\demoapp\tricore\tasking

2.3 SchM Files

File Name	AS403/ AS321	Description
SchM_<Mod>_<VendorId>_<ApiInfix>.h	Both	Implementation of SchM prototypes for SchM_Enter_<mod>_<VendorId>_<ApiInfix> and SchM_Exit_<mod>_<VendorId>_<ApiInfix> Location: \Aurix_MC-ISAR\integration_general\inc
SchM.c	Both	Sample Implementation of SchM_Enter_<Mod>_<VendorId>_<ApiInfix> and SchM_Exit_<Mod>_<VendorId>_<ApiInfix> functions

<Mod> refers to Adc, Icu, Wdg, Gpt, Can, Fls, FlsLoader, Lin and Spi

2.4 GNU Make Package

Package	AS403/ AS321	Description
GNU Make Files for Base Module	Both	Make File Package based on GNU. Location: \Aurix_MC-ISAR\mak

2.5 EcuM

File Name	AS403/ AS321	Description
EcuM.h	Both	Exports Driver functionality. Location: \Aurix_MC-ISAR\ecum_infineon_tricore\ssclinc
EcuM.c	Both	Call out of driver initializations Location: / Aurix_MC-ISAR\ecum_infineon_tricore\ssclsrc
EcuM_Callout_Stubs.c	Both	Call out of IRQ initializations. Location: \Aurix_MC-ISAR\ecum_infineon_tricore\ssclsrc
EcuM_Cbk.h	Both	Extern for driver declarations Location: \Aurix_MC-ISAR\ecum_infineon_tricore\ssclinc
Ecum.xdm	Both	Module parameter description (Tresos format) Location: \Aurix_MC-ISAR\tools\tresosECU\plugins\EcuM_Aurix\config
EcuM_Cfg.h	Both	Code Template Location: \Aurix_MC-ISAR\tools\tresosECU\plugins\EcuM_Aurix\generate\inc
EcuM_LCf.c	Both	Code Template Location:

		\\Aurix_MC-ISAR\tools\tresosECU\plugins\EcuM_Aurix\generate\src
EcuM_PBCfg.c	Both	Code Template Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\EcuM_Aurix\generate\src
EcuM.m	Both	Code Template Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\EcuM_Aurix\generate

2.6 IRQ

File Name	AS403/ AS321	Description
Irq.h	Both	Interrupt Priority Configuration File. Location: \\Aurix_MC-ISAR\irq_infineon_tricore\ssc\inc
Irq.c	Both	Interrupt Priority Initialization functions. Location: \\Aurix_MC-ISAR\irq_infineon_tricore\ssc\src
<Mod>_Irq.c	Both	Module specific interrupt service routines. Location: \\Aurix_MC-ISAR\irq_infineon_tricore\ssc\src
Irq.xdm	Both	Module parameter description (Tresos format) Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\Irq_Aurix\config
Irq.bmd	Both	Module parameter description Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\Irq_Aurix\autosar
anchors.xml	Both	This file contains reference to User Manual Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\Irq_Aurix
Irq_Cfg.h	Both	Code Template Location: \\Aurix_MC-ISAR\tools\tresosECU\plugins\Irq_Aurix\generate\inc
Aurix_MC-ISAR_UM_IRQDriver.pdf	Both	User Manual for IRQ driver
Mcal_Trap.c	Both	Trap handler Location: \\Aurix_MC-ISAR\irq_infineon_tricore\ssc\src
plugin.xml	Both	Tresos plugin file

<Mod> refers to Adc, Can, Ccu6, Dma, Eru, Eth, Fls, Gtm, Lin, Spi, Stm

3 Tool Information

Compiler Name	Tasking v4.2r2 Gnu v4.6.3.0
Processor Platform	TC233 AB PD TC234 AB PD TC234 AB ED TC237 AB PD
Evaluation Hardware	TriBoard TC2x3 TriBoard TC2x4 TriBoard TC2x7
Other Tools	NA
Code Configuration and Generation tool	EB Tresos Studio 16.0.0. Build-Nr: b141014-0350

TASKING v4.2r2:

Compiler Options:-

```
--core=tc1.6.x -t -Wa-gAHLs --emit-locals=-equ,-symbols -Wa-Ogs -Wa--error-limit=42 --iso=90 --eabi-compliant --integer-enumeration --language=-comments,-gcc,+volatile,-strings --switch=auto --align=0 --default-near-size=0 --default-a0-size=0 --default-a1-size=0 -O2ROPYGLKF-predict --tradeoff=4 -g --source -D_TASKING_C_TRICORE_=1
```

Mandatory Compiler Options:-

--core=tc1.6.x	Generate TriCore 1.6.x instructions in the assembly file
--eabi-compliant	Generated code needs to be completely EABI compliant.
--integer-enumeration	The enum-types are treated as integer
--language=-comments,-gcc,+volatile-strings	The gcc extensions are enabled, no optimization across volatile access, string literals are relaxed and '/' comments are not allowed.
--switch=auto	Choose the most efficient switch implementation based on code and data size and execution speed
--align=0	Minimum alignment default option selected
--default-near-size=0	Threshold for putting data in near filed is set to 0
--default-a0-size=0	Threshold for putting data in -a0 filed is set to 0
--default-a1-size=0	Threshold for putting data in -a1 filed is set to 0
-O2ROPYG	Mandatory compiler optimization level
--tradeoff=4	Compiler optimization is set to default for code size
-D_TASKING_C_TRICORE_=1	Defines the macro for TASKING compiler usage
-Wa-Ogs	Assembler Optimization option for instruction size and generic instructions

Optional Compiler Options:-

-Wa-gAHLs	Assembler Options for source line information and debug information.
--emit-locals=-equ,-symbols	Assembler option to disable Emit local EQU symbols and Emit local non-EQU symbols
-Wa--error-limit=42	Option to set the error limit
-g	Options to include DWARF debug information

--source	Option to merge C source code with generated assembly code in the output file
--eabi=+no-clear	Generate code to prevent uninitialized global/static variables from being cleared at program startup
--iso=90 / -iso=99	Compiler option ISO C standard-C90/C99
-OLKF-predict	Compiler optimization option to avoid compiler issues(TCVX-39721, TCVX-40081, TCVX-40088, TCVX-40256) Compiler optimization option -OK is added due to compiler issue TCVX-42331. This will be made as mandatory option from next release since it is a SIL4 compiler issue option and IFX release will not be tested without this option.

Linker Options:-

-D__CPU__= userdef16x_tc0 -Cmpe:tc0 -Otxycl --error-limit=42 -M -mcrfiklsmnoduq

Mandatory Linker Options:-

-Cmpe:tc0	Linker Option for single core
-D__CPU__= userdef16x_tc0	Option to define the CPU

Optional Linker Options:-

-Otxycl	Linker option for optimizations
--error-limit=42	Specifies the error limit
-M -mcrfiklsmnoduq	Specifies the options for the map file

Note: In general, there is no restriction on the usage of the linker options imposed by Mcal as the linker options do not influence the generated opcode. However the Mcal has only been tested with the specified Linker options.

GNU:

Compiler Options:-

-save-temps=obj -std=iso9899:1990 -ansi -fno-asm -ffreestanding -Wundef -Wp, -std=iso9899:1990 -fno-short-enums -fpeel-loops -falign-functions=4 -frecord-gcc-switches -fsection-anchors -funsigned-bitfields -ffunction-sections -fno-ivopts -fno-peekhole2 -nostartfiles -O3 -g3 -W -Wall -Wuninitialized -mtc161 -D_GNU_C_TRICORE_=1 -I "\$(PRODDIR)/tricore/include/machine"

Mandatory Compiler Options:-

-ansi	Disables features not compatible with ansi C
-ffreestanding	Built in functions handled more efficiently
-fno-short-enums	Enums are integer type
-fpeel-loops	Code optimization for loop peeling
-falign-functions=4	Starting address of functions are aligned to 4 byte boundary
-funsigned-bitfields	Bit fields to be used in the unsigned int data types.
-ffunction-sections	Functions normally placed in the ".text" are placed in their own sections.

-fno-ivopts	Perform induction variable optimizations (strength reduction, induction variable merging and induction variable elimination) on trees.
-fno-peekhole2	Disables RTL peekhole optimization after registers have been allocated but before scheduling.
-O3	Compiler Optimization level 3.
-mtc161	Compile code for specific CPU variant
-D_GNU_C_TRICORE_=1	Defines the macro for GNU compiler usage.
-I "\$(PRODDIR)/tricore/include/machine"	Path to include libraries where PRODDIR, defines the compiler installation path for the GNU tool

Optional Compiler Options:-

-std=iso9899:1990 / -std=iso9899:1999	Support all ISO C90/C99 standard
-Wp,-std=iso9899:1990 / -std=iso9899:1999	Preprocessor options not directly accessible via tricore-gcc
-save-temps=obj	Saves the obj files generated
-fno-asm	The asm keyword is disabled
-nostartfiles	Standard start-up files are not used. Startup files from MCAL are used.
-Wundef	Warn whenever an identifier which is not a macro is encountered
-frecord-gcc-switches	This switch causes the command line that was used to invoke the compiler to be recorded into the object file that is being created
-fsection-anchors	Option to reduce the number of symbolic address calculations by using shared "anchor" symbols to address nearby objects. This transformation can help to reduce the number of GOT entries and GOT accesses on some targets
-g3	Debug information is added to generated executable
-W	Enable additional warnings
-Wall	All warnings that occurred while compiling a program are displayed
-Wuninitialized	Warn if an automatic variable is used without first being initialized
-DTRIBOARD_TC23XX	Define the use of TRIBOARD

Linker Options:-

-nostartfiles -Wl,--allow-multiple-definition -Wl,--cref -Wl,--format=elf32-tricore -Wl,--mcpu=tc161 -Wl,--mem-holes -Wl,--extmap="a" -L "\$(PRODDIR)/tricore/include"
--

Mandatory Linker Options:-

-Wl,--mcpu=tc161	Option to check linking of objects with different cores
------------------	---

Optional Linker Options:-

-Wl,--mem-holes	Enables the spreading of data over different memory regions
-nostartfiles	The standard system startup files are not used when linking.
-Wl,--allow-multiple-definition	Allows multiple definitions of symbols
-Wl,--cref -Wl	Output a cross reference table
-Wl,--extmap="a"	Generates an extended map _le with additional information

Note: In general, there is no restriction on the usage of the linker options imposed by Mcal as the linker options do not influence the generated opcode. However the Mcal has only been tested with the specified Linker options.

4 Changes to Previous Version

4.1 Changes from V4.0.0 to V5.1.0

Package	Issue #	AS403 or AS321	Description	Modified Artifacts	
				Module	Files
BASE	0000051018-475	Both	Issue: Global variable used in EndInit protection API for password and timeout backup not implemented as redundant/diverse. Fix: Current timer reload value and the password are stored as diverse during the ResetENDINIT APIs execution and actual backup value is compared against the diverse value to identify the corruption in SetENDINIT APIs execution. If variable corruption is found, Mcal_SafeErrorHandler() function is called.	MCALLIB	Mcal_WdgLib.c
BASE	0000051018-191	Both	Issue: Comment for ErayPlI in clock calculator excel is wrong. No functional impact. Fix: Vco input frequency validation need to be included in Eray pll calculation. Also, allowed frequency in 23x device need to be corrected.	MCU	Aurix_MC-ISAR_MCU_Clock_Calculator.xlsm
BASE	0000051018-479	Both	Issue: Compiler errors seen when "ADC_DEINIT_API" is "STD_OFF". Fix: The pre-processor switch is included in the function which is invoked from Deinit function.	ADC	Adc_HwHandle.c
BASE	0000051018-253	Both	Issue : Timer tick generation for ADC hardware triggered conversion was not correct. Fix: GTM_Global_Numerator and GTM_Global_Denominator wrong swapping is corrected.	ADC	Adc.m
BASE	0000051018-487	Both	Issue: Memory section should be used only if module is being used. Fix: Module availability is checked before the memory sections are included.	GTM	Gtm.c
BASE	0000051018-465	Both	Issue: Safety error was reported when two channels are connected to same interrupt	GTM	Gtm.c

Package	Issue #	AS403 or AS321	Description	Modified Artifacts	
				Module	Files
			node; almost on same timings interrupts are produced. Fix: Removed the Safety error reporting, Interrupts generated will be checked against a valid source, but if it is not from any of the channels then it is not serviced.		
BASE	0000051018-459	Both	Issue: GTM TOM TGC configuration for internal trigger not correct. Fix: Changed the bit position for TGC register in Gtm.m to generate first element in "Gtm_kTomTcgConfig0[]" correctly for internal trigger control. Two bits are used to enable the channel and the value is now set correctly.	GTM	Gtm.m
BASE	0000051018-482	Both	Issue: Compiler through a warning, dead assignment to "LocalGptChannelPtr" eliminated in 'Gpt.c' file. Fix: Declaration of the variable has been corrected to avoid the warning.	GPT	Gpt.c
BASE	0000051018-265	Both	Issue: Re-entrancy statement in code differs from UM. Fix: Updated the comment in the two API description same as the comment in the User Manual.	PORT	Port.c Aurix_MC-ISAR_UM_PORTDriver.pdf
BASE	0000051018-197	Both	Issue: As per SAS_MCAL_SPI_9863, Separate DMA channels to be used for every kernel, resource clashes to be performed in configuration. Usage of same DMA channels is not validated in SPI. Fix: DMA channel usage is validated for uniqueness in configuration validation. The requirement ID is not SAS_MCAL_SPI_9863, as mentioned in JIRA name. It is traced to SAS_MCAL_SPI_9845.	SPI	Spi_Cfg.h
COM_BASIC	0000051018-158	Both	Issue: Since MoDataIndex1 not reset correctly in Can_IRxIsrExtractData, wrong messages are received for CAN message with high occurrence rate (1ms), while Rx is in Polling	CAN	Can_17_MCanP.c

Package	Issue #	AS403 or AS321	Description	Modified Artifacts	
				Module	Files
			mode, Fix: To resolve this issue, MoDataIndex1 variable reset is placed inside the loop for Can_IRxIsrExtractData and Can_IRxIsrExtractDataRuntime functions.		
COM_BASIC	0000051018-488	Both	Issue: GNU compiler reported warning as unused parameter for 'ControllerBaudrate' and unused variable for 'KerIndex' Fix: UNUSED_PARAMETER(KerIndex) is used to fix this issue.	CAN	Can_17_MCanP_Platform.c
COM_BASIC	0000051018-208	Both	Issue: SwHandles are wrong in Tx_Confirmation for Multiplexed transmission Fix: Swpdu handle should be stored without overwrite during multiplexed transmission request.	CAN	Can_17_MCanP_PBCfg.c
COM_BASIC	0000051018-204	Both	Issue: Consistency of CanHwControllerId and CanControllerId is not checked. Fix: An error check is done during generation to check the consistency of CanHwControllerId and CanControllerId.	CAN	Can_17_MCanP_PBCfg.c
COM_BASIC	0000051018-222	Both	Issue: "CanControllerFdBaudRate" cannot be set to 500Kbps. Fix: Configuration is modified to update range of CanControllerFdBaudRate to 100 - 5000.	CAN	Can_17_MCanP.xdm
COM_BASIC	0000051018-260	Both	Issue: CAN FD Frames are sent as CAN Frames (supports only Standard16Bit for Can_IdType). CANFD flag is not set correctly. Fix: According to AS4.1.3, implemented the support of Can_IdType which is Standard32Bit.	CAN	Can_17_MCanP.c

Package	Issue #	AS403 or AS321	Description	Modified Artifacts	
				Module	Files
COM_BASIC	0000051018-267	Both	Issue: The global variable "Can_SwObjectHandle" is defined with wrong array size. Fix: Removed CAN_TX_HARDWARE_OBJECTID_MAX macro and used CAN_TX_HARDWARE_MAX for the array index initialization.	CAN	Can_17_MCanP_Cfg.h
COM_BASIC	0000051018-436	Both	Issue: Wrong information is passed to upper layer in polling mode; issue is in index of loop, due to which ISR process all CAN MO instead of MO configured for FD mode. Fix: The for loop in Can_17_MCanP_IsrTransmitRuntime should be repeated for TxObjectcount/3 times for FD enabled controller and TxObjectcount times for FD disabled controller.	CAN	Can_17_MCanP.c
COM_BASIC	0000051018-282	Both	Issue: CAN_ISO_NUM_KERNEL macro is not generated for the devices which doesn't support ISO frame format. Fix: CAN_ISO_NUM_KERNEL macro used only if precompile macro CAN_HARDWARE_ISO_SUPPORT is STD_ON	CAN	Can_17_MCanP_Platform.c
COM_BASIC	0000051018-473	Both	Issue: Message object over-writing check inside function Can_IRxIsrExtractDataRuntime is implemented with wrong User Mode Access Macro to update the MOCTR SFR. Fix: Uses WRITE32 macro instead of MODIFY32 macro for the register access.	CAN	Can_17_MCanP.c
COM_BASIC	0000051018-469	Both	Issue: MO status (MSGLST, NEWDAT and RXPND) were not cleared by software. Because of this, after one FIFO circular Fill, the MO status will be MSGLST=DATA LOST. Fix: Clearing MO status (MSGLST, NEWDAT and RXPND) should be done after MSGLOST condition occurred during FIFO message reception.	CAN	Can_17_MCanP.c

Package	Issue #	AS403 or AS321	Description	Modified Artifacts	
				Module	Files
COM_BASIC	0000051018-440	Both	Issue: Wrong software pdu handle information passed to CAN-IF layer, if multiple MO object are configured as Multiplexed. Fix: Index of the array which stores software pdu handles are generated properly to resolve this issue.	CAN	Can_17_MCanP_PBCfg.c
COM_BASIC	0000051018-452	Both	Issue: Pending or unintended CAN Rx messages are not completely cleared when the CAN controller is set to STARTED state Fix: Can_IgnoreReceivedMsg function is updated to add a loop to clear all message objects for the respective controller in Interrupt mode and added function Can_IgnoreReceivedMsgRuntime to clear pending messages in polling mode.	CAN	Can_17_MCanP_Platform.c Can_17_MCanP.c Can_17_MCanP.h Can_17_MCanP_Platform.h
COM_BASIC	0000051018-490	Both	Issue: Compilation errors in Can_17_MCanP_PBCfg.c file Fix: Generation of " <i>Can_kTxObjectHthKer1Map</i> " corrected in template file	CAN	Can_17_MCanP_PBCfg.c
COM_BASIC	0000051018-448	Both	Issue: RSM tool reported a warning in CanTrcv_17_6251G_MainFunction Fix: To resolve this issue, void argument passed to the CanTrcv_17_6251G_MainFunction.	CANTRCV	CanTrcv_17_6251G.c
COM_BASIC	0000051018-444	Both	Issue: Main function is missing in CanTrcv_17_6250GV33 driver. Fix: Implemented CanTrcv_17_6250GV33_MainFunction as a dummy function in driver code.	CANTRCV	CanTrcv_17_6250GV33.c CanTrcv_17_6250GV33.h
COM_BASIC	0000051018-201	Both	Issue: CanTrcv_17_6250GV33.xdm file contains incorrect section. Fix: Removed wrong sections.	CANTRCV	CanTrcv_17_6250GV33.xdm

Package	Issue #	AS403 or AS321	Description	Modified Artifacts	
				Module	Files
MEM	0000051018-280	Both	Issue: GC not completed if erase suspend feature is active. Fix: Configured the default value for " <i>FlsEraseSuspendTimeout [6000]</i> " and also introduced a production error for suspend/resume timeout.	FLS	Fls_17_Pmu.c Fls_17_Pmu_Ver.c Fls_17_Pmu.h Fls_17_Pmu_Local.h Fls_17_Pmu.xdm Fls_17_Pmu_Cfg.h Fls_17_Pmu_PBCfg.c
COM_ENHANCED	0000051018-249	Both	Issue: API <i>Fr_17_Eray_TransmitTxLPdu()</i> returns E_OK, even input buffer is busy with some other transfer Fix: Corrected the return value.	FR	Fr_17_Eray.c
COM_ENHANCED	0000051018-190	Both	Issue: The value of <i>FrTimeoutDurationFactor</i> parameter was not proper. Fix: Default value of <i>FrTimeoutDurationFactor</i> is updated.	FR	Fr_17_Eray.xdm Aurix_MC-ISAR_UM_FR_17_ERAYDriver.pdf
COMPLEX	0000051018-143	Both	Issue: Memory section for " <i>FlsLoader_IGetFlashType</i> " function is wrong. Fix: Section changed for API " <i>extern uint32 FlsLoader_IGetFlashType(FlsLoader_AddressType TargetAddress)</i> " to FLSLOADER_START_SEC_WRITE_CODE	FLSLOADER	FlsLoader_Local.h
DEMOCD	0000051018-184	Both	Issue: Warnings observed during package testing. Fix: Macro name changed to <i>SENT_DRV_UNINITIALIZED</i> and <i>SENT_DRV_INITIALIZED</i> to remove warning from the WindRiver.	SENT	Sent.h Sent.c

Configuration changes

Compatibility check	Result
Is there any change in parameters supplied from previous version?	Yes
Added parameters	FLS : FLS_E_SUSPND_RESUME_TIMEOUT Eth : EthCtrlEnableCrcStripping
Deleted parameters	CANTRCV : CanTrcv_17_6250GV33ParameterDefinition
Modified parameters	Eth : EthCtrlRxBufLenByte and EthCtrlTxBufLenByte CAN : CanControllerFdBaudrateConfig FR : FrTimeoutDurationFactor
Can the previously saved configuration be reused?	Yes

File structure changes

Compatibility check	Result
Is there any change in file structure delivered from previous version?	None

4.2 Changes from Initial Version V0.1.0 to V4.0.0

Please refer below mentioned table for the details on previous versions (0.1.0, 0.1.1, 0.1.2, 1.0.0, 2.0.0, 3.0.0, 3.1.0 and 4.0.0) bug fixes.

Release No.	Release Version	Released Package zip name	Release Notes Name
REL-308	V4.0.0	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_BASE_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_BASE_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_COM_BASIC_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_COM_BASIC_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_COM_ENHANCED_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_COM_ENHANCED_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_CD_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_COMPLEX_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_DEMOCD_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_DEMOCD_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_MEM_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_MEM_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_STDLIN_COMPLEX_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PB_STDLIN_COMPLEX_Releasenotes.pdf
		MC-ISAR_AS3XX_AURIX_TC23X_AB_PC_LIB_V400_TASKING_V4_GNU.zip	MC-ISAR_AS3XX_AURIX_TC23X_AB_PC_LIB_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_BASE_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_BASE_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_COM_BASIC_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_COM_BASIC_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_COM_ENHANCED_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_COM_ENHANCED_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_CD_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_COMPLEX_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_DEMOCD_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_DEMOCD_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_MEM_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_MEM_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_STDLIN_COMPLEX_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PB_STDLIN_COMPLEX_Releasenotes.pdf
		MC-ISAR_AS4XX_AURIX_TC23X_AB_PC_LIB_V400_TASKING_V4_GNU.zip	MC-ISAR_AS4XX_AURIX_TC23X_AB_PC_LIB_Releasenotes.pdf

5 Installation Requirements

Refer AURIX_MC-ISAR_InstallationGuide.pdf on how to install the package.

Refer Demo_App.txt and ConfigurationReadMe.txt for details of the demo application.

To run “demoapp” for drivers, “BASE” package should exist.

Note:

Standalone STD_LIN use:

Mcu Tresos Plugin Files (applicable for NON-Autosar / Standalone usage only)

File Name	AS403/ AS321	Description
Mcu.xdm	Both	Module parameter description (Tresos format)
McuPreConfiguration.xdm	Both	Module Pre Configuration (Tresos format)
Mcu.bmd	Both	Module parameter description (Tresos format)
Mcu_Cfg.h	Both	Code Template
Mcu_PBCfg.c	Both	Code Template
Mcu_Bswmd.arxml	AS403	BSW Module description file
Mcu_Catalog.xml	AS403	Mapping of AUTOSAR-ENGINEERING-OBJECT to physical files
anchors.xml	Both	User Manual Reference file
plugin.xml	Both	Tresos plugin file
plugin.properties	Both	Tresos property file
MANIFEST.MF	Both	Tresos manifest file

Note: This plugins are intended for standalone StdLin driver usage (configuration generation) only. For autosar MCAL, the base package MCU plugins are used instead.

6 List of known Issues

GNU Compiler warning (GCC-238): bignum truncated to 4 bytes, compiler truncates value to 4 bytes and this warning has no effect on the generated code, so this warning can be ignored.

Module	Fault Id	AS403 or AS321	Description	Work Around
MCU	0000051018-697	Both	TRISTEN of standby and Wake-up Control Register (PMSWCR0) is not set, so Tristate enable bit(TRISTREQ)bit cannot be set. Tristate enable bit makes pads behave as inputs with weak pull-up or tristate on reset assertion/deassertion or Standby- Wake-up transition. So reset assertion/deassertion or Standby- Wake-up transition cannot be done tristate input.	None
MCU	0000051018-689	Both	API Mcu_ClrWkpStdbby and Mcu_SetStandbyCtrlReg are disabling the write protection of registers always. Whereas as per requirement when safety is enabled, MCU driver shall not explicitly disable write protection of the register and disabling of write protection shall be done by the user.	None
MCU	0000051018-539	Both	Various configuration parameters (McuErayPIIPDivider, McuErayPIINDivider, McuClockErayPIIFrequency etc) has EDITABLE, ENABLE and VISIBLE attributes set.	None
MCU	0000051018-683	Both	Hardware errata "PLL_TC.005" and "PLL_ERAY_TC.001" are not implemented as per workaround sequence suggested by hardware errata. Hardware errata suggested following sequence needs to be done before PLL initialization; however MCAL driver does this sequence after the PLL initialization. SCU_CCUCON0.B.CLKSEL = 0; /* switch system clock to another source different from PLL, e.g. back-up clock */ SCU_CCUCON0.B.UP = 1; /* request update */ SCU_PLLCON0.B.PLLPWD = 0; /* set PLL to power saving mode */ wait(10); /* wait 10µs */ SCU_PLLCON0.B.PLLPWD = 1; /* set PLL to normal behaviour */	Application can invoke the mentioned sequence (As in description field) before calling the Mcu_InitClock.
MCALLIB	0000051261-378	Both	Compiler warning is seen in "Mcal_Compiler.h" due to undefined preprocessing identifier "GNU_C_TRICORE".	None

Module	Fault Id	AS403 or AS321	Description	Work Around
ADC	0000051018-498	Both	The parameters AdcChannelLowLimit, AdcChannelHighLimit has 8/10/12 bit resolution. Based on the resolution these parameter can be following possible values 8bit : 0 - 255 10bit : 0 - 1023 12bit : 0 - 4095 Limit check is not enabled if user configuring value beyond these range.	User to ensure that valid values are configured as per resolution.
GPT	0000051018-526	Both	As per AS4XX, GptClockReferencePoint is not configurable, however current implementation allows the user to add multiple number of GptClockReferencePoint_0.	User to ensure that he doesn't configure GptClockReferencePoint for AS4XX.
GPT	0000051018-522	Both	GptEnableWakeup should be greyed out when GptReportWakeupSource is disabled.	User to ignore the configuration of GptEnableWakeup and GptWakeupConfiguration container when GptReportWakeupSource is disabled.
GTM	0000051261-377	Both	Configured value (i.e. for TIMi_IN_SRC) generated to route signal internally (without using external pins) from TIM to TOM, ATOM is not correct.	Use external connection to route signals from TIM to TOM, ATOM.
PWM	0000051018-517	Both	Types Pwm_ChannelIdentifierType and Pwm_GlobalChannelNumberType are present in header file but with vendor name prefix. However these are used by driver internally.	Integrator to ensure that name collision doesn't happen by following the vendor prefix in name of parameters which he/she defines.
DIO	0000051018-687	AS321	DIO xdm marks the module as precompile, whereas design marks it as linktime.	None
DIO	0000051018-588	Both	APIs "Dio_MaskedWritePort" and "Dio_WritePort " are not doing the range check on input parameters.	User has to pass the value to API based on pin availability in specific port.
DIO	0000051018-532	Both	TC23x (e.g. TC234, TC237) have 42 ports; however the value generated for DIO_MAX_VALID_PORT_PIN_ID is based on 41 ports.	None.
ICU	0000051018-566	Both	Variable "ChlType" defined in function "Icu_17_GtmCcu6_Isr" can be unused if precompile macro "ICU_TIM_SIGMEAS_USED" is set to "STD_ON" but precompile macros "ICU_GET_INPUT_STATE_API", "STD_OFF", "ICU_GET_TIME_ELAPSED_API" are set to STD_OFF.	None

Module	Fault Id	AS403 or AS321	Description	Work Around
ICU	0000051018-496	Both	Max Range Check has not performed in API: Icu_17_GtmCcu6_EnableMultiEdgeDetection for EdgeCount Parameter	User to ensure that he/she doesn't pass value greater than "2^24-1".
ICU	0000051018-519	Both	Spurious interrupt is generated if TIM is configured in 'timestamp mode' and enabling, disabling of notification is done as depicted below : <pre> Fun() { Disable notification() /*do some calculations */ Enable notification() } </pre>	Following steps need to be taken care by application before and after the call of "Enable Notification" : 1. Disable SRE bit of the used TIM Channel 2. Call enable Notification function 3. Clear the pending interrupt of the particular channel using CLRR bit 4. Re-enable SRE of the Used TIM Channel
SPI	0000051018-109	Both	If multiple slaves present in one QSPI kernel and some has CS as active high. After the module initialization, when job to another slave connected to this QSPI module is started. Then slave with CS high active will also get the MRST, MTSR and Clock as valid signal.	1. CS for slaves can be driven by GPIO. 2. Use all slave devices having same CS polarity i.e. either Active High or Low. 3. Partition active High or Low slaves in different QSPI kernel.
SPI	0000051018-115	Both	SPI Sequence is not getting transmitted when configured buffer size is 1 and SPI drive is configured for async mode POLLING and SPI_LEVEL_DELIVERED is 2.	Configure buffer size greater than 1.
SPI	0000051018-137	Both	Static variable "CurHwUsed" defined in Spi_SyncTransmit is not under mem-section.	Add linker command in script to keep variable 'CurHwUsed' in desired section as per product requirement.
SPI	0000051018-126	Both	In ASYNC transmit API, Queue availability can be corrupted due to lack of semaphore protection.	Use static task scheduling with non-preemptive mode i.e. SPI driver is handled by single task.
FR	0000051018-144	Both	Buffer reconfiguration array error due to two different tresos configuration nodes are considered while generation.	If the unused frames (not referred from LPdu) in the FrIfFrameTriggering is removed. Then the Flextrain tool will not modify the generated Fr_17_Eray_PBCfg.c file and provide a message "LPDU index to message buffer mapping is NOT required".

Module	Fault Id	AS403 or AS321	Description	Work Around
STDLIN	0000051018-187	Both	An ASCLIN module configured as LIN slave node could report a wrong collision detection error during reception of LIN header after detecting a first correct collision detection error during the transmission of a response field of the previous LIN frame.	None
IOM	0000051018-98	Both	If a clock divider value RMC > 1 is selected in register IOM_CLC, more than one write access may be performed to the IOM register address space within one IOM clock cycle.	Set IOM_CLC.RMC = 1 when configuring the IOM.
FR	0000051018-100	Both	FIFO depth is not calculated correctly. FIFO index should start after Dynamic, Static messages. FIFO Message Count = Total_Message – (Static + Dynamic) Message. However FIFO Index calculation doesn't take care of this calculation.	FrFifoDepth should be configured to total number of FIFO LPDUs. i.e. FrFifoDepth=No.Fifo LPDUs
SENT	0000051018-180	Both	First Write Access to Endinit-protected Registers after ENDINIT Status Change	None

7 HIS-MISRA Violations

PC-Lint (Ver 9.00k) configuration files used for MISRA evaluation is available at Installation Docs location

MISRA violations due to SFR access / compiler intrinsic functions

MISRA_2004_Rule	Rule Description	Global Deviation	Applicable Modules
8.1	Functions shall have prototype declarations and the prototype shall be visible at both the function definition and call.	Violation when used compiler specific Intrinsic functions via compiler abstraction file/s (e.g. IMASKLDMST, EXTRACT)	All modules where such a scenario exists
17.4	Array indexing shall be the only allowed form of pointer arithmetic	Deviation Reasons: 1. Possibility of Memory wastage Ex. To overcome MISRA violation, we need to mention possible Maximum Channels configured and all PB config sets. Whenever lesser channels configured in any PB configuration, we need to mention worst case possible MAXIMUM CHANNELS available for array size. 2. SFR access 3. Autosar API's shall pass buffer pointers and wherein we need to increment the pointer. This MISRA will not allow doing pointer arithmetic or pointer increment option. But the ECR (ECR_196) actually allows pointer increment/decrement.	1) All module with variant PB configuration 2) All Memory modules
18.4	Unions shall not be used.	Unions are used in SFR file for register description	All modules
11.1 11.3	A cast should not be performed between a pointer type and an integral type. From MISRA document: Casting between a pointer and an integer type should be avoided where possible, but may be unavoidable when addressing memory mapped registers or other hardware specific features.	Allowed only if this issue is because of Sfr file. For e.g. in the Sfr file, the following definition leads to this MISRA #define DSADC_ACCPROT ((*(volatile Ifx_DSADC_ACCPROT*) 0xF0024090u)).	All modules where such a scenario exists

MISRA violations due to AUTOSAR framework

MISRA_2004_Rule	Rule Description	Global Deviation	Applicable Modules
19.15	Precautions shall be taken in order to prevent the contents of a header file being included twice	Deviation shall be taken only when Memmap.h is included in the source files (Note : Memmap.h file is imported according to AutoSAR requirement)	
19.1	#include statements in a file should only be preceded by other pre-processor directives or comments	Deviation shall be taken only when Memmap.h is included in the source files. Structure of Memmap.h according to Autosar is such that it would have to be included in different areas in the code file.	All Modules
8.10	All declarations and definitions of objects or functions at file scope shall have internal linkage unless external linkage is required.	Debug variables can't be made as local if debug support is enabled. Hence declared as global	All modules with debug variable support
19.4	C macros shall only expand to a braced initializer, a constant, a string literal, a parenthesized expression, a type qualifier, a storage class specifier, or a do-while-zero construct	Deviation is taken only for <Mod>_GetVersionInfo API	All Modules

MISRA violations in module

MISRA_2004_Rule	Rule Description	Global Deviation	Applicable Modules
19.7	A function should be used in preference to a function-like macro	Function like macros can be used in the following scenario: a) GetVersionInfo API implemented as Macro b) #error directly for disabled API's (i.e. API's which can be switched off by a configuration switch are re-defined as a macro to give error if called). c) Used in configuration to improve readability	All Modules
19.13	The # and ## operators should not be used	# used only to set the number of bits required for IMASK instruction in Tasking compiler specific assembly code. There is no impact of order of evaluation with the # operator at this intrinsic function.	MCALLIB
13.7	Boolean operations whose results are invariant shall not be permitted.	MISRA states to check only for logical operators but PC LINT also checks for bitwise operators. Confirmed as a tool issue from PCLINT.	All modules where such a scenario exists
11.5	A cast shall not be performed that removes any const or volatile qualification from the type addressed by a pointer.	Cannot be avoided in case of parameters for Mcal_SetAtomic. This is also solvable but gives a compile time warning in case of windriver compiler.	ADC,PWM,GTM, SPI,ICU,MCALLIB,GPT
5.1	Identifiers (internal and external) shall not rely on the significance of more than 31 characters.	Cannot be avoided in case of MemMap section names and enumerated name of configuration.	ADC, WDG
8.5	There shall be no definitions of objects or functions in a header file	Deviation is taken for extern inline functions.	All Modules
8.8	An external object or function shall be declared in one and only one file	Tool issue: states the violation also for extern variable declarations in header file	All Modules
16.7	A pointer parameter in a function prototype should be declared as pointer to const if the pointer is not used to modify the addressed object	ICU: Deviation is taken for the pointer parameter in a function, which is modified via assembly instructions. PWM: Though function parameter is only	ICU, PWM

		read in the function, 'const' specifier is not used, to help user define different value at runtime.	
1.2	No reliance shall be placed on undefined or unspecified behavior	Configuration and runtime logic ensures that, unspecified behavior does not occur.	PWM
14.2	All non-null statements shall either: (a) have at least one side-effect however executed, or (b) cause control flow to change.	Syscall will generate Trap Class 6, hence not a null statement	MCALLIB
15.5	Every switch statement shall have at least one case clause.	Switch statement does not have case clause, if only Request Source 0 (RS0) alone used in the driver configuration. Otherwise, the violation doesn't exist.	ADC
8.7	Objects shall be defined at block scope if they are only accessed from within a single function	The variables are made global to optimize the code generation and to reduce the code complexity.	ADC, DIO, MCU, GPT, ICU, MCALLIB, PWM
5.6	No identifier in one name space should have the same spelling as an identifier in another name space, with the exception of structure member and union member names.	Deviation taken for tag name and structure type name, while defining structures.	ALL

8 List of Limitations and Deviations

Customer shall consider the disclaimers mentioned in page 1

8.1 List of Deviations to AUTOSAR specifications

The following table summarizes the list of deviations to AUTOSAR SWS

Module	Req ID	Description
MCU	MCU213	Smu_Irq.c shall have the interrupt service routine function for clock failure.
MCU	MCU215	Mcu_LCfg.c shall not be available as it is not required.
ADC	ADC289	The ADC module's priority mechanism allows the configuration of 255 priority levels (0...254). Since the priority level 255 is set to Hw trigger groups in ADC_PRIORITY_HW_SW mechanism.
ADC	ADC369	The group channel result values are stored in the order of channels configured in AdcGroupDefinition parameter, not in the ascending channel number order.
ADC	ADC417	In ADC_PRIORITY_HW_SW mechanism, If more Groups has same priority level and enters the HW SW implemented Queue, the implementation shall schedule the channel group conversion based on the group id. i.e., the higher group id's are serviced first.
ADC	ADC430	The ADC module does not support suspend/resume mechanism since there is no hardware support.
ADC	ADC019_Conf	Adc Channel resolution configuration is moved to Global and Kernel Input class container and a reference parameter is provided to select one of 4 input classes.
ADC	ADC088_Conf	The functionality of AdcPrescale is achieved through the parameter AdcAnalogClockDivider
ADC	ADC287_Conf	Since the priority level 255 is set to Hw trigger groups in ADC_PRIORITY_HW_SW mechanism, the range of the parameter AdcGroupPriority is limited to 255 priority levels (0...254).
ADC	ADC290_Conf	The functionality of AdcChannelSampTime is achieved through the parameter AdcGlobChSampleTime and AdcKernelChSampleTime
ADC	ADC316_Conf	The AdcStreamingBufferMode is configured even in Single Access mode.
ADC	ADC401_Conf	Since there is no hardware support, the requirement ADC401_Conf (AdcHwTrigTimer) is achieved through parameter AdcGtmTimerPeriod.
ADC	ADC435_Conf	The range (ADC_GROUP_REPL_SUSPEND_RESUME) of the parameter AdcGroupReplacement is not supported since there is no hardware support.
PWM	PWM094	Hardware does not support to identify production errors; hence the Dem_EventIdType is not supported.
PWM	PWM104	Hardware does not support to identify production errors, hence Dem_ReportErrorStatus is not supported.
PWM	PWM136_Conf	Multiplicity of PwmChannelClass parameter is set to 1 irrespective of DET switch, since the Pwm channel class configuration is important for the driver implementation even if DET is OFF. Also PWM_FIXED_PERIOD_CENTER_ALIGNED range is added along with the other Autosar ranges as microcontroller specific feature.

ICU	ICU119	Resetting GTM related Interrupt flag is handled by GTM driver. ERU related Interrupt flag is handled by ICU driver.
ICU	ICU213	No Dem reported for ICU, hence Dem_ReportErrorStatus will not be used.
ICU	ICU229	Clearing Interrupt flag related to GTM is handled by GTM Driver. ERU related Interrupt Flags are cleared in ICU driver.
ICU	ICU247	ICU driver is implemented as Post Build and hence Icu_Lcfg.c file will not be provided.
ICU	ICU251	Icu_17_GetVersionInfo Api is implemented as a Macro and moved to Icu_17.h. So the inclusion of Det.h is required in Icu_17.h. Hence instead of Icu_17.c, Icu_17.h includes Det.h.
ICU	ICU281	GTM related clock and Hardware dependent properties are handled in GTM driver. The used TIM or ERU channel number alone is mentioned in ConfigType.
ICU	ICU038	GTM related clock and Hardware dependent properties are handled in GTM driver. The used TIM or ERU channel number is mentioned in ConfigType. Also Wakeup support is mentioned.
ICU	ICU251, ICU252	These 2 requirements are not implemented. Please refer Bugzilla 52388 for details.
GPT	GPT375	Gpt_GetVersionInfo API is implemented as a Macro and moved to Gpt.h. So the inclusion of Det.h is required in Gpt.h. Hence instead of Gpt.c, Gpt.h includes Det.h.
GPT	GPT157	The function Gpt_DisableWakeup is only feasible, if GptReportWakeupSource is statically configured available. This requirement ignored based on bugzilla number 17946.
GPT	GPT158	The function Gpt_EnableWakeup is only feasible, if GptReportWakeupSource is statically configured available. This requirement ignored based on bugzilla number 17946.
GPT	GPT255	The function Gpt_SetMode is only available if the configuration parameter GptReportWakeupSource is enabled. This requirement ignored based on bugzilla number 17946.
ICU	ICU295 (AS4.0.3)	ICU_DUTY_CYCLE and its related functionalities are not provided. A discussion is ongoing on this in Bugzilla 52843
CAN	CAN435, CAN436, CAN441, CAN442	This version of CAN driver does not support multiple main functions for read and write. Ref UTP: AI00068021.
CAN	CAN192 CAN431_Conf, CAN229	Since all the CAN controller state transitions are synchronous operations, the API Can_InitController() is implemented as a synchronous function and the free running timer provided by Os.h is not used by the CAN driver .
CAN	CAN196, CAN197	Can_SetControllerMode will not explicitly deal with interrupts, as the hardware settings done in the respective modes, ensures the expected interrupt behavior.
CAN	CAN069, CAN215, CAN278, CAN285, CAN287, CAN288, CAN286, CAN399, CAN400, CAN215, CAN434	Transmit cancellation and Identical Id cancellation features are not supported

CAN	SWS_Can_00416 - Autosar 4.1.2 AI00237562	Represents the Identifier of an L-PDU. The two most significant bits specify the frame type. As per original requirement 30 th bit of Extended ID and 14 th bit of standard ID is set always as zero .After implementing CAN FD these bits are used for representing the CAN FD frames by setting that bit to one. When both standard and extended IDs are configured datatype of Can_IdType used is uint32 and this case also for standard ID we are setting 14 th bit for CAN FD frames.
CAN	ECUC_Can_00473 - Autosar 4.1.2 AI00237562	This optional container contains bit timing related configuration parameters of the CAN controller(s) for payload and CRC of a CAN FD frame. If this container exists the controller supports CAN FD frames. This container added as part of CAN FD implementation. CAN FD baud rate which allows the user to transmit upto 64 bytes of data at a higher data rate with valid FD data length as 0 – 8, 12, 16, 20, 24, 32, 48 and 64.
CAN	ECUC_Can_00481 - Autosar 4.1.2 AI00237562	Specifies the data segment baud rate of the controller in kbps when the data is transmitted in CAN FD mode. This parameter added as part of CAN FD implementation for setting baud rate of data segment.
CAN	ECUC_Can_00476 - Autosar 4.1.2 AI00237562	Specifies propagation delay in time quantas when data is transmitted in CAN FD. This parameter added as part of CAN FD implementation.
CAN	ECUC_Can_00477 - Autosar 4.1.2 AI00237562	Specifies phase segment 1 in time quantas when data is transmitted in CAN FD. This parameter added as part of CAN FD implementation.
CAN	ECUC_Can_00478 - Autosar 4.1.2 AI00237562	Specifies phase segment 2 in time quantas when data is transmitted in CAN FD. This parameter added as part of CAN FD implementation.
CAN	ECUC_Can_00479 - Autosar 4.1.2 AI00237562	Specifies the synchronization jump width for the controller in time quantas quantas when data is transmitted in CAN FD. This parameter added as part of CAN FD implementation.
CAN	ECUC_Can_00480 - Autosar 4.1.2 AI00237562	Specifies the Transceiver Delay Compensation Offset in ns. If not specified Transceiver Delay Compensation is disabled. This parameter added as part of CAN FD implementation.
CAN	ECUC_Can_00475 - Autosar 4.1.2 AI00237562	Specifies if the bit rate switching shall be used for transmissions. If FALSE: CAN FD frames shall be sent without bit rate switching. This parameter added as part of CAN FD implementation.
CAN	SWS_CAN_00486 - Autosar 4.1.2 AI00237562	The CAN Frame has to be sent according to the two most significant bits of Can_PduType->id. The CAN FD frame bit is only evaluated if FD baud rate is active. This requirement is added as part of CAN FD implementation.
CAN	Bugzilla – RfC 56415 AI00237562	CAN FD utilize a second baud rate for payload transmission to increase communication throughput. This requirement is added as part of CAN FD implementation. Additional FD baud rate container added along with CAN387_Conf requirement.

CAN	AI00237562	The maximum size of transmit/receive PDUs and PDU buffers throughout the CAN Stack have to be extended to a maximum size of 64 bytes if CAN FD is available. This requirement is added as part of CAN FD implementation. This is a deviation from CAN218 requirement
CAN	CAN218	If development error detection for the Can module is enabled: The function Can_Write shall raise the error CAN_E_PARAM_DLC and shall return CAN_NOT_OK if the length is more than 8 byte This requirement has been deviated due to the CAN FD implementation. If CAN FD is enabled this check is done for invalid length. Valid Data Length for CAN FD is 0-8,12,16,20,24,32,48, 64. Any length other than specified valid of length , the function Can_Write shall raise the error CAN_E_PARAM_DLC and shall return CAN_NOT_OK
Flexray	NA	Since FlexRay Interface XDM cannot be modified (It is not deliverable from Infineon), following solution is assumed in case of Single Slot Status Error: If user wants to configure FrIfDemFTSlotStatusRef to report DEM for slot status information, this parameter should be configured in first element(Index 0) of FrIfFrameTriggering container and should be ignored in rest of the instances of FrIfFrameTriggering container. Refer Bugzilla 51695
Flexray	NA	Config parameter FrIfAlwaysTransmit is not related to hardware feature transmission mode single shot / continuous. The driver always configures a message buffer for single shot mode, if it is transmit message buffer. Refer Bugzilla 29048
Flexray	NA	If FrIf has multiple post-build configurations and multiple cluster configurations a post-build configuration, then there is possibility that one FrController is referenced multiple times through FrIfFrCtrlRef. In this case, the first FrIfFrCtrlRef instance, encountered in the search, is considered for cluster specific configuration parameters and LPDU configuration parameters. Refer Bugzilla 28924
Flexray	FR576	If (Fr_ListSize > 15), new DET, FR_17_ERAY_MAX_LIST_SIZE is introduced. If Fr_ListSize is greater than 15, size of Fr_ListSize id set to 15, as maximum of only 15 sync frames have to be supported. Refer Bugzilla 52410
Flexray	FR514	According to the specification the enumeration FR_ChannelType should start with 0, but FR_ChannelType is defined to be starting with 1 due to the optimization for accessing SUCC1 register. Refer Bugzilla 29300
Flexray	Fr_ReceiveRxLpdu	Fr_ReceiveRxLpdu : SWS does not mention the FIFO overrun Error. In case of FIFO overrun error, E_NOT_OK is returned and *Fr_LPduStatusPtr will hold FR_RECEIVED. Refer Bugzilla 51693
Flexray	FrDemCtrlTestResultRef	The production error FrDemCtrlTestResultRef for Hardware access is not supported. Refer Bugzilla: 29729
Ethernet	ETH005_Conf	Configuration parameter EthVersionInfoApiMacro has been disabled by default and not used.because to maintain uniform across all modules, driver uses EthVersionInfoApi parameter
Ethernet	ETH002_Conf	EthMaxCtrlsSupported parameter is disabled for configuration because all Aurix variants have single Ethernet MAC
Ethernet	ETH010_Conf, ETH011_Conf, ETH007_Conf, ETH012_Conf	Parameters EthCtrlEnableRxInterrupt, EthCtrlEnableTxInterrupt, EthCtrlIdx, EthCtrlEnableMii are available in the EthGeneral container. These are not part of EthConfigSet.

FlsLoader	NA	Instance ID is not returned by FlsLoader_GetVersionInfo in case of Autosar V3.2, as it is not significant in case of complex driver.
FlsLoader	NA	Hardware verification feature for Flash programming/erase is currently not supported.
FlsLoader	NA	The following features are not within the scope of Driver and hence not supported: 1. Interrupt handling for single bit errors detected during read access. 2. Double bit error handling. 3. Margin control for changing the margin levels for read operations to find problematic array bits in flash. 4. VER bit checking is not supported. 5. HSM protection support.
FLS	FLS048, FLS279_Conf	Protection setting is not used in the FLS Driver as it is best handled by the FlsLoader Driver and hence the parameter FlsProtection is not supported.
FLS	FLS277_Conf, FLS278_Conf	FlsMaxWriteFastMode and FlsMaxWriteNormalMode are not supported as Write is performed for 8/32 bytes depending on the data size and page start address. So Fls_17_SetMode does not have any impact for Fls_17_Write.
FLS	FLS362	Timeout monitoring for Read and Compare is not supported since Read time depends on FSI clock and considerably small to handle the timeout using FlsMainFunction scheduling time. The SWS also suggests in FLS362 as an open point that if expected Read/Compare times are not detailed enough then read/compare job should be out of timeout supervision.
FEE	FEE021	For read requests (Fee_Read / Fee_17_GetPrevData): Due to the usage of the double-sector algorithm for EEPROM emulation, address of the logical block will not be calculated to locate the block in memory, instead a look-up table called "cache-table" will be used.
FEE	FEE023	A read request will fail with MEMIF_BLOCK_INCONSISTENT in the following cases: - The logical block does not exist. - The most recent version of the block is not complete (assuming block number was identified without read error). - At least one Flash read access resulted in an error (e.g. when an uncorrectable ECC error was detected).
FEE	FEE037	A block is invalidated by writing appropriate infrastructure data of the block to DFlash (DF_EEPROM).
FEE	FEE024 & FEE036	For write requests: Due to the usage of the double-sector algorithm, address of the logical block is not calculated. Instead the block will be written to DFlash (DF_EEPROM) as per the double-sector algorithm.
FEE	FEE080	If Fee_Cancel API is called when a user requested read job is ongoing, then an IFX FEE specific API Fls_17_CancelNonEraseJobs will be called, which cancels only an ongoing read job.

FEE	FEE009, FEE080 and FEE081	An ongoing write operation requested by the user will not be affected by Fee_Cancel. Blocks containing immediate data cannot be written instantaneously (in conflict with FEE009). Fls_17_Cancel will not be called (in conflict with FEE080). Fee_Cancel will not reset the FEE Module's internal variables (in conflict with FEE081).
FEE	Extension of Fee_Cancel	In case that a user request is pending but is not yet being executed, the request can be canceled; Fee_Cancel will only reset the FEE Module's internal variables. During garbage collection, the status will switch back from MEMIF_BUSY to MEMIF_BUSY_INTERNAL when the garbage collection is still running or to MEMIF_IDLE when it has finished.
FEE	Extension of FEE090	Fee_GetStatus will return MEMIF_BUSY not only while the user job is being executed but also while it is accepted but waiting for execution (i.e., while internal management operation is going on). During garbage collection, the status MEMIF_BUSY_INTERNAL will be returned only before a user job is accepted; as soon as a user job is accepted, the status will switch to MEMIF_BUSY.
FEE	FEE067	Since double-sector algorithm is used with threshold limit for triggering sector change, immediate block's write requests (during sector change) can be accommodated within the pre-erased threshold area of the active sector. Hence, the API Fee_EraseImmediateBlock is implemented as a dummy function.
FEE	Extension of Fee_Cancel	Fee_Cancel API will report the error FEE_E_INVALID_CANCEL to DET, if there is no job ongoing to be cancelled. This is in line with AS4.0.3 requirements
FEE	FEE010	The development error FEE_E_BUSY_INTERNAL is not supported (except for Fee_SetMode API) as the user request will be accepted by the FEE driver (but will be executed with delay) even when the FEE driver status is MEMIF_BUSY_INTERNAL.
FEE	Extension	To allow the usage of FEE driver across applications, few configuration parameters have been implemented as Post Build parameters.

8.2 Functional Limitations

The following table summarizes the most important functional limitations due to Hardware feature and Software design constraints. For more details refer to the user manuals of the individual module.

Module	Limitation
ADC	- Maximum number of groups can be 32 in each ADC HW Unit - Maximum number of channel configurations can be 256 in each ADC HW Unit - A group can have up to 8 channels. - Interrupt latency/execution time problems - There may be errors if interrupt is not handled fast enough. - ADC HW unit configured for faster conversion time: ADC Group configured for Software trigger, Continuous conversion mode may complete the conversion before completion interrupt service routine. In this scenario, the program execution control will always be in ISR. The next statement of the application will never get the chance for execution. - The above scenario can even happen for Groups using Timer trigger source and external request trigger source. In case of Timer trigger source, execution control will never come out of ISR when timer period is small and ADC HW is triggered and the conversion finishes before the completion of ISR. In case of external request trigger source, execution control will never come out of ISR when external requesting is triggering the conversion and A/D conversion is finished before the ISR execution is completed. - Any two ADC hardware units can be enabled for using external multiplexer. - Group with EMUX SCAN enabled - AdcGroupDefination should contain one channel with Emux select line configured - Group should be configured for continuous conversion mode - The channel under EMUX SCAN Group should have EMUX select value greater than 0. - When a Group with EMUX SCAN enabled is requested for conversion, then the other Group with EMUX SCAN enabled is not accepted for conversion in any priority mechanism except No Priority Queue mechanism (DET: ADC_E_BUSY will be raised) - Hardware triggered groups are not supported for EMUX SCAN feature. - EMUX SCAN enabled group can not have Limit checking enabled channel. - In EMUX SCAN mode, Notification function is called only if the conversion for streaming number of samples is completed. - When a Software triggered group with RS2 (Background Request Source) is requested for conversion, then the other Software/Hardware triggered group with RS2 is not accepted for conversion in any priority mechanism, irrespective of the Adc channel group kernels (DET: ADC_E_BUSY will be raised). - When a Hardware triggered group with RS2 (Background Request Source) is requested for conversion, then the other Software triggered group with RS2 is not accepted for conversion in any priority mechanism, irrespective of the Adc channel group kernels (DET: ADC_E_BUSY will be raised). - When Hardware triggered group with RS2 (Background Request Source) is requested for conversion, then the other Hardware triggered group with RS2 of the same ADC kernel should not be called for conversion. - When Hardware triggered groups with RS2 (Background Request Source) from different ADC kernels are requested for conversion, the Request source expect only the trigger line of the first group which is enabled (Adc_EnableHardwareTrigger) for conversion. User should start the trigger by calling Adc_17_StartGtmTimer () or Adc_17_SetEruTrigger () APIs. - When Hardware triggered groups with RS2 (Background Request Source) from different ADC kernels are requested to stop the conversion, the Request source is stopped only after the last group is disabled (Adc_DisableHardwareTrigger). User should stop the trigger by calling Adc_17_StopGtmTimer () or Adc_17_ResetEruTrigger () APIs. - The Group notification depends on RS priority as well as the interrupt node priority (does not always follow the priorities of the groups). - All the API execution should be from lower priority as compared to ADC ISRs. - Hardware trigger Groups of an ADC kernel which shares the same trigger unit (Eg: ERU

	trigger) should not be enabled for the conversion or wait for the trigger at the same time. 21. When a limit check enabled group is requested (start/enable) for a conversion, another limit check enabled group of the same ADC kernel should not be requested for the conversion. 22. API <code>Adc_17_GetStartupCalStatus ()</code> returns <code>E_NOT_OK</code>, if the configured <code>AdcConfigSets</code> does not have identical ADC kernels. E.g.: if <code>AdcConfigSet_0</code> has <code>ADC_HW_UNIT0</code> & <code>ADC_HW_UNIT1</code>, <code>AdcConfigSet_1</code> has <code>ADC_HW_UNIT0</code> & <code>ADC_HW_UNIT2</code>, then <code>Adc_17_GetStartupCalStatus ()</code> always returns <code>E_NOT_OK</code> for both the <code>AdcConfigSets</code>.
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ADC	Software Limitation in ADC_PRIORITY_HW mechanism: 1. Groups with the same request source AdcGroupRequestSource will have the same priority level as per request source priority. 2. There can be only one request by the same request source Group. In case if there is a Second request by the group with same request source, a DET: ADC_E_BUSY will be reported. 3. ADC Groups configured for different conversion request sources but sharing some of the ADC internal channels cannot be started simultaneously. Here we are overruling the priority mechanism. In such scenario, Start/Enable service will report DET: ADC_E_BUSY. <u>For example,</u> Consider two ADC Groups configured as: AdcGroup0-RS2, Group Definition {15, 10} AdcGroup1-RS1, Group Definition {5, 15} AdcGroup0 and AdcGroup1 cannot be requested for conversion simultaneously. If AdcGroup0 is requested for conversion and then immediately request for conversion for AdcGroup1 is received, AdcGroup1 is not serviced, instead a DET: ADC_E_BUSY will be reported even if its request source priority level is higher than AdcGroup0 request source priority level. Here we are overruling the priority mechanism because of internal channel sharing by the requested groups.
ADC	Software Limitation in ADC_PRIORITY_HW_SW mechanism: 1. ADC Groups sharing some of the ADC internal channels cannot be enabled for conversion and will not enter the Software implemented Queue. Here we are overruling the priority mechanism. In such scenario, Start/Enable service will report DET: ADC_E_BUSY. <u>For example,</u> Consider two ADC Groups configured as: AdcGroup0-RS2 – Priority 0, Group Definition {15, 10} AdcGroup1-RS0 – Priority 2, Group Definition {5, 15} AdcGroup0 and AdcGroup1 cannot be requested for conversion simultaneously. If AdcGroup0 is requested for conversion and then immediately request for conversion for AdcGroup1 is received, AdcGroup1 is not serviced, instead a DET: ADC_E_BUSY will be reported even if its Group priority level is higher than AdcGroup0 trigger source priority level. Here we are overruling the priority mechanism because of internal channel sharing by the requested groups.
ADC	Software Configuration Limitation: 1. AdcChannelId for Adc Channels should be in a continuous sequence i.e. (0, 1, 2, 3...) 2. AdcGroupId for ADC Groups should be in a continuous sequence i.e. (0, 1, 2, 3...) 3. AdcGroupId for ADC Groups are generated by the configuration tool. User has to refer the AdcGroupId by their symbolic names. The symbolic names are configurable by the user. 4. ADC Groups are sorted in ADC Group configuration structure in the order of their trigger type and request source starting from software trigger groups of RS0 to RS2 and then followed by hardware trigger groups of RS0 to RS2. The configuration tool will automatically arrange the Groups in the order of their AdcGroupRequestSource 5. The first element of the Group definition should be the number of configured channels in the group. From Second element will give the channel ID. <u>For example,</u> A group with channels (say) {4, 6, 7, 3, and 1}; first element of Group definition is the total number of channels which is 5.
ADC	Software Limitation in ADC_NON_AUTOSAR Mode: 1. Priority mechanism is disabled. The driver is always in ADC_PRIORITY_NONE with Queue disabled.

	2. The Group channels have to be arranged in the descending order of their analog channel numbers. 3. EMUX Scan cannot be used. 4. Limit checking cannot be used.
ADC	Software Limitation in Safety feature: 1. Background source can't be used for any of the group. 2. Emux Scan can't be used for any of the group. 3. Non Autosar result handling mode can't be used.
PWM	Software Configuration Limitation: 1. User has to take care for sequential allocation of TOM/ATOM Channel resources for PWM Grouping. 2. The port pin allocation should be done in such a way that it does not conflict with other allocations.
SPI	SPI202_Conf: The configuration restricts the range of parameter SpiDataWidth to 2...32, this is in accordance with the hardware specification.
MCU	If McuK2DivSteps is configured with a higher value for smaller deltas in the frequency then there is a possibility of redundant jumps to the same frequency as the result of lesser granularity (for SCU_PLLCON1.B. K2DIV). User has a choice to reduce the number of steps to avoid redundant jumps. However, redundant jumps do not cause any harm in achieving the final frequency.
LIN	The Dem configuration parameter LIN_E_TIMEOUT is not applicable for Aurix family microcontrollers as none of registers required to be waited for Timeout period. Hence the Dem api Dem_ReportErrorStatus() is not invoked in the LIN Driver
LIN	Since System Clock is common to all ASCLIN channels, instead "LinClockRef {LIN_CLOCK_SRC_REFERENCE}" the channel dependent configuration parameter, the new configuration parameter [LinSysClockRef] shall be used for the same functionality as part of LinGeneral container
CAN	When the CAN driver calls the callback function CanIf_RxIndication upon FIFO reception event, upper layer has to ensure that the control is returned back before next FIFO threshold is reached, to avoid data loss.
CAN	MultiCAN+ Module has option to use fCLC or Oscillator clock or E-Ray PLL Clock to calculate Baud rate and Bit timing operations. Current implementation supports only fCLC or E-Ray PLL Clock for baud rate and bit timing calculations.
CAN	Multiplexed transmission feature does not always ensure the transmission of frames based on priority. Since it depends on the instant at which Can_17_MCanP_Write() is called and the instant at which MutliCAN+ actually carries out the transmit acceptance filtering.
Flexray	Driver supports only 10Mbps baudrate in accordance with FlexRay protocol spec v2.1
Flexray	A maximum of 255 (Id = 0 to 254) LPDUs can be configured
Flexray	Only following oscillator frequencies are supported: 16Mhz, 20MHz and 24MHz
FLS	In case of Fls_17_Pmu_Write the SourceAddressPtr containing the content to be written should be even aligned.
FEE	In order to avoid the search for the location of the latest copy of the block for each of the block operations, a lookup table called "cache table" is prepared, which contains the address and status information of all the blocks. Hence the first read/write/invalidate request after Fee_Init, might have to wait until the cache table is prepared.
FEE	Interrupted programming of the last page of a logical block (weakly programmed data block marker page – a non-zero value or un-correctable ECC error) will be determined as a consistent block. But during the next initialization (cache table build), if that weakly programmed page's content is read as 0's (without un-correctable ECC error – i.e., it looks like a missing data block marker page) then the block will be considered as inconsistent.
FEE	Value of the configuration parameter FeeMaxBlockCount should be appropriate. Note that the overestimated value of this parameter would result in RAM space wastage, as this value

	determines the size of the cache table for configured and un-configured blocks..
FEE	If FEE finds an all 0 DF_EEPROM (both half of the FEE sectors are erased), then FEE does not erase the DF_EEPROM again. The assumption is that the DF_EEPROM is erased correctly. Possible incompletely erased DF_EEPROM, might later (while programming can) lead to FEE reaching an illegal state. This condition is not handled by the FEE driver.
STDLIN	I_ifc_rx, I_ifc_tx : The AscLin hardware handles the data in terms of complete frames and not character by character. Hence these functions are not supported
STDLIN	I_ifc_aux: The AscLin hardware handles the data in terms of complete frames including the break and sync detection. Hence this api is not supported
STDLIN	I_ifc_ioctl: There are no protocol and interface specific parameters to be controlled and hence this api is not supported
STDLIN	Id_read_by_id_callout: api is not supported as it is optional.
STDLIN	Static prototypes of the apis are not supported as it is optional
CRC	While calculating the CRC with 16 bit hardware the input stream is expected to be aligned to 16bit word boundary. If not aligned the remaining 8 bits are assumed to be zero.
CRC	While calculating the CRC with 32 bit hardware the input stream is expected to be aligned to 32bit word boundary. If not aligned the remaining bits are assumed to be zero.

9 Sub-Derivative Specification

This section explains the supported sub-derivatives. The lists of supported sub-derivatives are:

1. SAK-TC233L-16F200F
2. SAK-TC233L-32F200F
3. SAK-TC233LC-24F133F
4. SAK-TC233LP-16F200F
5. SAK-TC233LP-32F200F
6. SAK-TC234L-32F200F
7. SAK-TC234LA-32F200F
8. SAK-TC234LP-16F200F
9. SAK-TC234LP-32F200F
10. SAK-TC234LX-32F200F
11. SAK-TC237L-32F200S
12. SAK-TC237LP-32F200S
13. SAL-TC233L-32F200F
14. SAL-TC233LP-16F200F
15. SAL-TC233LP-32F200F
16. SAL-TC234L-32F200F
17. SAL-TC234LP-16F200F
18. SAL-TC234LP-32F200F
19. SAL-TC237L-32F200S
20. SAL-TC237LP-32F200S

10 SafeTLib Compatibility

Table 1 SafeTLib Package information

Derivative	Release date	Deliverable Details
TC23xAB	20-Jan-2016	Product Name : PRO-SIL_AURIX_SafeTlib_Extended Release Version : Rel652 (PR5) Installer Name : PRO-SIL_AURIX_SAFETLIB_EXTENDED_TC23XAB_AS403_PR5_Rel652.exe* PRO-SIL_AURIX_SAFETLIB_EXTENDED_TC23XAB_AS321_PR5_Rel652.exe* *Build_Mcal_SafeTlib.bat should be taken from the MC_ISAR_TC23XAB_REL_334_SafeTlib_REL_652.zip for the MCAL and SafeTlib Demo Integration.
TC23xAB	11-July-2016	Product Name : PRO-SIL_AURIX_SafeTlib_Extended Release Version : Rel631 (MR1) Installer Name : PRO-SIL_AURIX_SAFETLIB_EXTENDED_TC23XAB_AS403_MR1_Rel631.exe* PRO-SIL_AURIX_SAFETLIB_EXTENDED_TC23XAB_AS403_MR1_Rel631.exe* *Build_Mcal_SafeTlib.bat should be taken from the MC_ISAR_TC23XAB_REL_334_SafeTlib_REL_631.zip for the MCAL and SafeTlib Demo Integration.
TC23xAB	17-Oct-2016	Product Name : PRO-SIL_AURIX_SafeTlib_Extended Release Version : Rel632 (MR2) Installer Name : PRO-SIL_AURIX_SAFETLIB_EXTENDED_TC23X_AS403_MR2_Rel632.exe PRO-SIL_AURIX_SAFETLIB_EXTENDED_TC23X_AS403_MR2_Rel632.exe

Note: Hint for SafeTlib User Mode

1. SBST Runtime Test is not supported in User Mode
2. Following APIs needs to be executed before switching to User Mode
 SI_SwitchTstPhase(); (From Test Handler, this function is used to switch SafeTlib to run phase)
 SafeWdgIf_Init(); (Use: Initialization of internal watchdog).